Exercise list R8b — Executive compensation and managerial incentives

Worked solutions

Statements are in `Listas/lista-r8b-executive-compensation.md`; each exercise is cited by number and printed page and its full text is reproduced right before its solution. Short answers: `lista-r8b-executive-compensation-solutions-short.pdf`. Editions are the local copies: Ross = Ross, Westerfield, Jordan & Lamb, *Fundamentos de Administração Financeira*, 9th ed. (2013); Berk = Berk, DeMarzo & Harford, *Fundamentals of Corporate Finance*, 2nd ed. (2011); Grinblatt–Titman (G&T), *Financial Markets and Corporate Strategy*, 2nd ed. (2002); Higgins = Koski, Higgins & Mitton, *Analysis for Financial Management*, 13th ed. (2022); Amaro de Matos, *Theoretical Foundations of Corporate Finance* (2001); Tirole, *The Theory of Corporate Finance* (2005). In the Tirole exercises everyone is risk neutral, the rate of interest is 0, the entrepreneur is protected by limited liability and $\Delta p \equiv p_H - p_L$. Amounts are in millions of dollars unless stated. Where a book gives no numbers, an illustrative parameter point is chosen and labelled as such; the conclusions do not depend on it.

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1. Is CEO pay too high? — Ross CQ 1.16, p. 23

Statement.

Source: Ross, Cap. 1, Questão 16 (Revisão de conceitos e questões instigantes), p. 23 (PDF 47) · Answer in book: no

16. Remuneração dos executivos [OA3] Os analistas têm indicado que a remuneração dos administradores de primeiro escalão nos Estados Unidos é alta demais e deve ser reduzida. Por exemplo, concentrando-nos nas grandes corporações, Ray Irani, da Occidental Petroleum, é um dos CEOs mais bem pagos, tendo ganhado cerca de \$54,4 milhões em 2007 e \$550 milhões no período de 2003 a 2007. Tais valores são excessivos? Ao responder essa pergunta, seria importante reconhecer que superestrelas dos esportes, como Tiger Woods, grandes celebridades, como Tom Hanks e Oprah Winfrey, e muitos outros profissionais de sucesso em suas respectivas áreas ganham tanto quanto ou mais.

What is asked. Ray Irani (Occidental Petroleum) earned about \$54.4m in 2007 and \$550m over 2003–2007. Are such amounts excessive, given that sports stars and celebrities earn as much or more?

Approach. The manager–shareholder agency problem (Ross Cap. 1, §1.4, "Remuneração dos administradores") says pay is a tool for aligning managers with shareholders. The right test is not the *level* of pay but (i) whether it was set at arm's length in a competitive market and (ii) how strongly it moves with the value the CEO creates (pay-for-performance sensitivity, G&T §18.5), net of luck the CEO does not control (Result 18.9).

Notation.

- W: CEO pay in a year; ΔV: change in shareholder value attributable to the CEO.
- s: pay-for-performance sensitivity, dollars of CEO wealth per \$1,000 of shareholder value.

Step 1 — Put the numbers on a common scale. The five-year total is spread over five years:

$$\frac{550}{5} = 110 \text{ per year}, \qquad \frac{54.4}{110} = 0.4945, \tag{1}$$

so 2007 was about half of the period average. A level of pay means little without the size of the firm: \$54.4m is 1% of a value increase of

$$\frac{54.4}{0.01} = 5{,}440, \tag{2}$$

i.e. \$5.44bn, a modest fraction of the value of a large oil company.

Step 2 — The superstar argument. Ross's own examples (p. 14) put CEO pay beside other stars: Ellison (\$193m), Rowling (\$300m), Winfrey (\$275m), Woods (\$115m). Where a scarce talent is applied to a very large base (a global audience, or a firm worth tens of billions), small differences in ability are worth a lot, so competition for that talent pushes pay up. On this view large CEO pay is a market price, not excess.

Step 3 — Where the analogy breaks. Three differences matter for the agency view:

- *Who sets the pay.* A sponsor or studio pays a star with its own money at arm's length. A CEO's pay is set by a board that the CEO often influences — the agency problem itself (G&T §18.1; Tirole §7.2.1.2 on boards "in cahoots" with management).
- *Whose output is measured.* A golfer's score is his own. Occidental's value moves with the oil price, which no CEO controls; pay tied to the raw stock price rewards luck (G&T Result 18.9).
- *Sensitivity, not level.* Jensen and Murphy (G&T p. 645) found that a CEO who buys a \$10m jet loses only about \$30,000:

$$s = \frac{30{,}000}{10{,}000{,}000} \times 1{,}000 = 3 \text{ per } \$1{,}000. \tag{3}$$

  Their complaint was that pay was *too insensitive* to performance, not too high.

Step 4 — The test. Pay is excessive when it is not explained by value created relative to peers: when it rises with the industry tide (oil prices) as much as with relative performance, when it is protected on the downside (repriced options, large severance after failure), or when the board is not independent. It is not excessive merely because the number is large.

Answer . Not necessarily. \$54.4m (and \$110m a year on average over 2003–2007) can be a fair price for scarce talent in a very large firm, as with sports and entertainment stars. It is excessive if it was not set at arm's length or if it pays for industry-wide luck (the oil price) rather than value created relative to peers; the evidence (Jensen–Murphy, \$3 per \$1,000) points to weak sensitivity rather than to levels as the main problem.

Intuition. Shareholders should care about the slope of pay with respect to value created, not the intercept. A high level with a steep, luck-filtered slope can be cheap; a modest level with no slope buys nothing.

Common mistakes.

- Judging pay by its level alone, without the firm's size or the value created.
- Accepting the superstar analogy uncritically: stars' pay is set at arm's length, CEOs' pay by boards they influence.
- Ignoring that an oil company's value is driven largely by the oil price, outside the CEO's control.

Book answer. No answer in the book.

Ref: Ross Cap. 1, §1.4 "O problema de agência e o controle da empresa", "Remuneração dos administradores", pp. 13–14; G&T §18.5 "Is Executive Pay Closely Tied to Performance?", pp. 645–647, Result 18.9, p. 645.

2. A small CEO stake and a value-destroying acquisition — Berk Ch. 22 Problem 3, p. 677

Statement.

Source: Berk, Ch. 22, Problem 3, p. 677 (PDF 708) · Answer in book: no

3. You are invested in GreenFrame, Inc. The CEO owns 3% of GreenFrame and is considering an acquisition. If the acquisition destroys \$50 million of GreenFrame's value, but the present value of the CEO's compensation increases by \$5 million, will he be better or worse off?

Context needed to solve it *From Berk §22.3 "Reasons to Acquire", "Managerial Motives to Merge", p. 659 (PDF 690):*

Conflicts of Interest. Managers may prefer to run a larger company due to the additional pay and prestige it brings. Because most CEOs hold only a small fraction of their firm's stock, they may not bear enough of the cost of an otherwise bad merger that increases their personal benefits. For example, a CEO who owns 1% of her firm's stock bears 1% of every dollar lost on a bad acquisition, but enjoys 100% of the gains in compensation and prestige that come with being the CEO of a larger company. If the acquisition destroys \$100 million in shareholder value, but increases the present value of her compensation by more than \$1 million, she will prefer to execute the merger anyway. Why would the board of directors create these incentives? Either due to poor monitoring of the manager, or belief that the strategy is correct even if the stock market disagrees, boards typically increase the pay of CEOs along with the size of the firm, even if the size comes at the expense of poorly performing acquisitions.

What is asked. The CEO owns 3% of GreenFrame. An acquisition destroys \$50m of value but raises the present value of his pay by \$5m. Is he better or worse off?

Approach. Berk's "Conflicts of Interest" passage (§22.3, p. 659): the CEO bears only his ownership share of value lost but keeps 100% of the pay increase. Compare the two.

Notation.

- $\alpha = 0.03$: CEO's ownership fraction; $\Delta V = -50$: change in firm value; $\Delta W = 5$: change in PV of his pay.

Step 1 — His share of the value destroyed. His shares lose

$$\alpha\,|\Delta V| = (0.03)(50) = 1.5. \tag{4}$$

Step 2 — Net effect on his wealth.

$$\Delta W - \alpha|\Delta V| = (5) - (1.5) = 3.5 > 0. \tag{5}$$

He is \$3.5m better off.

Step 3 — When would he refuse?. He refuses only if $\alpha|\Delta V| > \Delta W$, i.e. if his stake exceeded

$$\alpha^* = \frac{\Delta W}{|\Delta V|} = \frac{(5)}{(50)} = 0.10, \tag{6}$$

or, at 3%, if the acquisition destroyed more than $5/0.03 = 166.7$ of value. Berk's own example has the same structure: at 1% ownership a \$100m loss costs her $0.01 \times 100 = 1$, less than a pay rise of more than \$1m.

Answer . Better off by $5 - 0.03 \times 50 =$ \$3.5m. With a 3% stake he would accept any deal destroying up to \$166.7m of value for a \$5m pay rise.

Intuition. Size-linked pay is a private benefit of growth. With a small stake the CEO internalizes almost none of the cost, so pay that rises with firm size, not with value, rewards empire building.

Common mistakes.

- Comparing \$5m with the whole \$50m: he bears only 3% of it.
- Forgetting that the pay increase is itself a cost to shareholders: their total loss is larger than \$50m unless it is already included.

Book answer. No answer in the book.

Ref: Berk §22.3 "Reasons to Acquire", "Managerial Motives to Merge — Conflicts of Interest", p. 659.

3. Why managers want growth — Grinblatt & Titman 18.1, p. 653

Statement.

Source: Grinblatt, Ch. 18, Exercise 18.1, p. 653 (PDF 666) · Answer in book: no

18.1. Discuss why managers might tend to want their organizations to grow.

What is asked. List and explain the reasons why managers may push their firms to grow, even beyond the value-maximizing size.

Approach. G&T §18.1 and §18.3 (Result 18.4): managers bias investment toward size, diversification and their own expertise because their pay, prestige, job security and private benefits rise with size while they bear only a small fraction of the cost (Jensen–Meckling; compare Exercise 2).

Notation.

- α: manager's ownership share; NPV < 0: value lost on an expansion; ΔW: increase in pay; b: private benefits (prestige, perquisites, security).

Step 1 — The decision rule of a self-interested manager. He expands iff

$$\Delta W + b > \alpha\,|\text{NPV}|. \tag{7}$$

With α small the right side is small, so even clearly negative-NPV growth passes. Jensen–Meckling's example (G&T p. 630): at 5% ownership a \$1m jet costs the manager $0.05 \times 1{,}000{,}000 = 50{,}000$.

Step 2 — Why the left side grows with size.

- *Pay*: compensation consultants use the size of the organization as a key input (p. 638); larger firms justify higher salaries.
- *Prestige and power*: running a larger company carries status and more perquisites.
- *Job security*: a larger, more diversified firm is less likely to get into financial trouble. Only 43% of CEOs keep their jobs after bankruptcy (Gilson, p. 637). Managers cannot diversify their human capital, so they value diversifiable risk reduction that shareholders do not.
- *Careers and loyalty*: growth creates promotion opportunities for the managers and for employees they feel loyal to (stakeholder view, p. 628).
- *Entrenchment*: investing in assets that fit the manager's expertise makes him harder to replace (Shleifer–Vishny, p. 637).

Step 3 — Free cash flow makes it possible. Growth financed from retained earnings avoids the scrutiny of the capital market. RJR Nabisco's baking unit planned to spend \$2.8bn for \$148m of annual savings (p. 638):

$$\frac{148}{2{,}800} = 0.0529, \tag{8}$$

a pretax return of about 5%, scaled back after the LBO cut the cash available.

Step 4 — Remedies. Tie pay to value rather than size (stock-based pay, or earnings net of a capital charge: G&T p. 649 warns that pay on total earnings "will provide an incentive to increase the scale of the corporation's operations"); raise managerial ownership; use debt to drain free cash flow (Result 18.7); a monitoring board or blockholder.

Answer . Managers want growth because pay, prestige, perquisites, career opportunities and job security all increase with size and diversification, while their small ownership stake makes them bear little of the cost of negative-NPV expansion; free cash flow lets them finance it without outside scrutiny.

Intuition. Growth is a private good for managers and a public bad for diffuse shareholders whenever it has negative NPV; pay linked to size subsidizes it.

Common mistakes.

- Treating all growth as bad: the problem is growth with negative NPV.
- Forgetting the risk motive: managers value diversification that shareholders can get for free.

Book answer. No answer in the book.

Ref: G&T Ch. 18, introduction and §18.1, pp. 628–630; §18.3 "The Investment Choices Managers Prefer", pp. 636–638, Result 18.4, p. 638; p. 649.

4. A bonus after a boom year — Higgins 1.2, p. 32

Statement.

Source: Higgins, Ch. 1, Exercise 1.2, p. 32 (PDF 53) · Answer in book: no (even-numbered; answers only in the instructor resources)

2. Braintree Corporation, a biotechnology company, has \$7 billion in assets, \$5 billion in equity, and earned a profit of \$100 million last year as the economy boomed. Senior management proposes paying themselves a large cash bonus in recognition of their performance. As a member of Braintree's board of directors, how would you respond to this proposal?

Context needed to solve it *From Higgins Ch. 1, "Economic Income vs. Accounting Income — Imputed Costs", pp. 27–29 (PDF 48–50):*

A similar but subtler problem exists on the cost side of the income statement. It involves the cost of equity capital. Polaris's accountants acknowledge that in 2020 the company had use of $1,434 million of shareholders' money, measured at book value. They would further acknowledge that Polaris could not have operated without this money and that this money is not free. Just as creditors earn interest on loans, equity investors expect a return on their investments. [. . .]

[. . .] Observe that while the accountant shows earnings of $414 million, the economist records a profit of only $271 million. These numbers differ because the economist includes a $143 million charge as a cost of equity capital, while the accountant pretends equity is free. (We will consider ways to estimate a company's cost of equity capital in Chapter 8. Here, for illustrative purposes only, I have assumed a 10 percent annual equity cost and applied it to the book value of Polaris's equity [$143 million = 10% × $1,434 million].)

($ millions)	Accountant	Economist
Operating income	$592	$592
Interest expense	67	67
Other nonoperating expenses	4	4
Cost of equity		143
Income before taxes	521	378
Provision for taxes	107	107
Accounting earnings	$414	
Economic earnings		$271

The distinction between accounting earnings and economic earnings might be only a curiosity if everyone understood that positive accounting earnings are not necessarily a sign of superior or even commendable performance. But when many labor unions and politicians view accounting profits as evidence that a company can afford higher wages, higher taxes, or more onerous regulation, and when most managements view such profits as justification for distributing handsome performance bonuses, the distinction can be an important one. [. . .] Remember too that a company is not shooting par unless its economic profits are zero or greater.

What is asked. Braintree has $7bn of assets, $5bn of equity and earned $100m in a boom year. Management asks for a large cash bonus. How should a director respond?

Approach. Higgins's "Imputed Costs" (pp. 27–29): accounting profit ignores the cost of equity, so a positive profit is not evidence of good performance. Add the agency lesson that pay should not reward luck (G&T Result 18.9).

Notation.

- $NI = 100$: net income; $E = 5{,}000$: book equity; $TA = 7{,}000$: assets; k_E: cost of equity (Higgins's illustrative 10%).

Step 1 — Returns.

$$\text{ROE} = \frac{NI}{E} = \frac{(100)}{(5{,}000)} = 0.02, \tag{9}$$

$$\text{ROA} = \frac{NI}{TA} = \frac{(100)}{(7{,}000)} = 0.0143. \tag{10}$$

Step 2 — Economic profit. Charge equity at Higgins's illustrative $k_E = 10\%$:

$$k_E E = (0.10)(5{,}000) = 500, \tag{11}$$

$$NI - k_E E = (100) - (500) = -400. \tag{12}$$

Shareholders earned $400m less than their capital could have earned elsewhere; break-even would need a profit of $500m, five times what was earned.

Step 3 — Luck versus performance. The profit came "as the economy boomed". A common shock that lifts all firms is extraneous risk; rewarding it pays for luck. Performance should be

measured relative to peers in the same boom (relative performance, Result 18.10). In biotechnology, annual accounting profit is also a poor measure of value created (R&D is expensed, payoffs are lumpy), another reason not to pay on it.

Step 4 — The board's response. Decline the bonus as proposed. Offer instead a plan that (i) pays on economic profit (net of the cost of equity) or on stock-price performance, (ii) relative to industry peers, (iii) deferred or in restricted stock, so that pay follows value created over several years.

Answer . Reject it. ROE is only 2% (ROA 1.4%); with a 10% cost of equity the economic profit is $100 - 500 = -\$400$m, and the profit reflects a booming economy rather than management's contribution. Any bonus should be tied to economic profit relative to peers.

Intuition. Equity is not free. A company "is not shooting par unless its economic profits are zero or greater" (Higgins, p. 29), and a boom that lifts everyone is not managerial performance.

Common mistakes.

- Treating any positive accounting profit as success.
- Comparing ROE with zero instead of with the cost of equity.
- Rewarding the economy's performance instead of the managers'.

Book answer. No answer in the book (even-numbered problem).

Ref: Higgins Ch. 1, "Economic Income vs. Accounting Income — Imputed Costs", pp. 27–29; G&T Results 18.9–18.10, pp. 645–649.

5. Stock-based pay and market-to-book — Grinblatt & Titman 18.7, p. 653

Statement.

Source: Grinblatt, Ch. 18, Exercise 18.7, p. 653 (PDF 666) · Answer in book: no

18.7. The tendency of firms to use stock-based compensation is higher for firms with higher market-to-book ratios. Provide two explanations for this empirical observation.

What is asked. Give two explanations for why firms with higher market-to-book ratios use more stock-based compensation.

Approach. A correlation has two directions of causality. Either stock-based pay *raises* value (incentives), or high market-to-book firms *choose* stock-based pay because of what they are (growth options, discretion, poor accounting measures) or because their managers know something (signalling). G&T §18.5.

Step 1 — Explanation 1: incentives raise value. Pay tied to the stock price aligns managers with shareholders, so they cut waste and pick positive-NPV projects; market value rises relative to book. Evidence: stock prices rose about 20% around the adoption of performance plans (Tehranian–Waegelein); Mehran (1995) finds market-to-book positively related to performance-based pay (p. 647).

Step 2 — Explanation 2: growth firms optimally choose stock pay. A high market-to-book ratio signals intangible assets and growth options (p. 636). In such firms (i) managers need discretion and are hard to monitor (Result 18.5: discretion is more valuable in uncertain environments); (ii) current accounting earnings say little about the value being created by growth options, so the stock price is the best available performance measure (Result 18.11); (iii) cash-constrained growth firms substitute equity for cash salary. Equity and options are used more in growth firms (Clinch; Smith–Watts; Gaver–Gaver, footnote 17).

Step 3 — A third channel: information. Managers accept stock-based pay more readily when they know the stock is undervalued (Example 18.5). With options struck at \$20, Jack, who believes his stock is worth \$30, values each option at

$$\max(0,\, 30 - 20) = 10, \tag{13}$$

while Peggy, who believes \$15, values it at $\max(0, 15 - 20) = 0$. Adoption reveals good news and the price rises even with no change in effort. Managers are also more willing to adopt such plans after prices have risen.

Answer . (1) Causation from pay to value: stock-based pay improves incentives and raises market value relative to book. (2) Reverse causation through firm type: high market-to-book firms are growth firms with intangible assets, high managerial discretion and poor accounting performance measures, for which stock-based pay is the efficient contract. (A third: managers with favourable private information accept stock-based pay, so adoption signals good news.)

Intuition. The stock price is the best measure of performance exactly where the book value is the worst measure of the firm, so the two go together even without any incentive effect.

Common mistakes.

- Reading the correlation as proof that stock pay creates value.
- Giving two versions of the same incentive story instead of two distinct mechanisms.

Book answer. No answer in the book.

Ref: G&T §18.2, p. 636; §18.5 "How Does Firm Value Relate to the Use of Performance-Based Pay?", pp. 646–648, Example 18.5; Results 18.5 and 18.11, pp. 639, 650.

6. Paying the Chicago Bulls' coach — Grinblatt & Titman 18.9, p. 654

Statement.

Source: Grinblatt, Ch. 18, Exercise 18.9, p. 654 (PDF 667) · Answer in book: no

18.9. Suppose that you are designing the compensation contract for the Chicago Bulls' new coach. Two main alternatives are possible. In (a) you will design his bonus based on the total number of wins during the season and the team's success during the playoffs and will ignore any specific decisions made by the coach. In (b) you will consider the specific measures taken by the coach and, perhaps with the help of independent outside experts, will base the compensation on the quality of those decisions but will ignore the number of wins during the season. Explain the advantages and disadvantages of the two compensation contracts.

What is asked. Compare a bonus on wins and playoff success (output) with pay based on experts' assessment of the coach's decisions (input), giving the advantages and disadvantages of each.

Approach. G&T §18.5 "Measuring Inputs versus Measuring Outputs" and the tenant-farmer trade-off: output pay aligns the agent with the principal's goal but loads him with risk he cannot control; input measures filter out luck but are costly, subjective and hard to contract on. Tirole's incentive constraint makes the cost of a noisy measure precise: the bonus must be at least B/Δ, where Δ is how much the measure responds to the agent's behaviour.

Notation.

- Coach behaves (good decisions) or shirks (saves effort, favours players he likes): private benefit B if he shirks.
- Signal $s \in \{\text{wins, expert rating}\}$; π_H^s, π_L^s: probability of a good signal if he behaves / shirks; $\Delta_s = \pi_H^s - \pi_L^s$.
- b_s: bonus paid on a good signal; the coach is protected by limited liability (no negative pay).

Step 1 — Incentive constraint. He behaves iff $\pi_H^s b_s \geq \pi_L^s b_s + B$, i.e.

$$b_s \geq \frac{B}{\Delta_s}, \tag{14}$$

and the expected bonus paid is $\pi_H^s B/\Delta_s$.

Step 2 — Illustration (numbers chosen for illustration). Let $B = 1$. Wins depend heavily on the roster, injuries and luck: $\pi_H = 0.6$, $\pi_L = 0.4$. A careful expert review is more responsive to the coach's choices: $\pi_H = 0.9$, $\pi_L = 0.3$.

$$\text{wins:} \quad b = \frac{(1)}{(0.6 - 0.4)} = 5, \qquad \text{expected pay} = (0.6)(5) = 3, \tag{15}$$

$$\text{experts:} \quad b = \frac{(1)}{(0.9 - 0.3)} = 1.667, \qquad \text{expected pay} = (0.9)(1.667) = 1.5. \tag{16}$$

The more informative measure buys the same behaviour for half the cost. With a risk-averse coach the gap is larger still, because the noisy contract also needs a risk premium.

Step 3 — (a) Bonus on wins and playoffs. Advantages: it rewards exactly what the owner wants (wins, playoff revenue); it is objective, verifiable and cheap to administer; it lets the coach use his own expertise without second-guessing; it attracts coaches confident in their ability. Disadvantages: wins depend on factors the coach does not control (the roster he inherits, injuries, opponents, luck), so he bears extraneous risk (Result 18.9) and must be paid a premium, or is paid for luck; it invites short-termism (playing injured stars, neglecting young players' development, trading the future for this season); a rebuilding team gives a weak link between effort and wins.

Step 4 — (b) Pay on the quality of decisions. Advantages: it filters out luck and roster quality, so risk and the required bonus fall (Step 2); it can reward long-term decisions (player development) that wins this season do not show. Disadvantages: evaluation is costly and subjective, hence hard to enforce (non-verifiable); experts lack the coach's information about players and opponents, so the scheme rewards conventional, easily defended decisions and discourages good unconventional ones; the coach may spend effort pleasing the evaluators (influence activities); and it ignores the outcome the owner ultimately cares about.

Step 5 — A better design. Combine the two: a bonus on wins *relative to expectations or to comparable teams* (relative performance, Result 18.10) plus a discretionary component for decisions; weight (b) more for a rebuilding team, (a) more for a contender.

Answer . (a) Output-based: aligned with the owner's goal, objective and cheap, respects the coach's information; but exposes him to luck and roster risk and encourages short-termism. (b) Input-based: filters luck and can reward long-term decisions, needing a smaller bonus for the same incentives; but costly, subjective, non-verifiable and biased toward conventional decisions. A mix with wins measured relative to a benchmark is best.

Intuition. Pay should load on the measure that best separates good from bad behaviour, Δ_s in (14); outcomes are the principal's true objective but often a noisy signal of the agent's choices.

Common mistakes.

- Calling output pay "obviously right" because it targets what the owner wants: its cost is the risk the coach cannot control.
- Ignoring that input measures are not verifiable and can be gamed by pleasing the evaluators.

Book answer. No answer in the book.

Ref: G&T §18.5 "The Agency Problem", "Measuring Inputs versus Measuring Outputs", "Designing Optimal Incentive Contracts", pp. 643–645, Results 18.9–18.10; Tirole §3.2.2, (IC_b), pp. 116–117; Amaro de Matos §3.1.2, p. 70.

7. CEO contracts at Chrysler and Chevron — Grinblatt & Titman 18.6, p. 653

Statement.

Source: Grinblatt, Ch. 18, Exercise 18.6, p. 653 (PDF 666) · Answer in book: no

18.6. You are a member of the compensation committee of the board of directors for both Chrysler and Chevron. How should the compensation contracts for the CEOs of these two companies differ?

What is asked. As a compensation committee member of both boards, explain how the CEO contracts at an automaker (Chrysler) and an oil company (Chevron) should differ.

Approach. Three pieces of G&T §18.5: remove extraneous risk (Result 18.9), relative performance has an advantage and a competitive side effect (Result 18.10), and stock- versus cash-flow-based pay (Result 18.11); plus the free-cash-flow overinvestment problem of mature, cash-rich firms (§18.3, the oil industry example).

Step 1 — Chevron: filter the oil price. Chevron's value is driven to a large extent by the oil price, which is observable, common to all oil producers and outside the CEO's control. Pay on the raw stock price loads the CEO with that risk. Indexing options or bonuses to the oil price or to a peer group of oil companies removes it. Illustration (assumed numbers): if the oil factor explains $R^2 = 0.7$ of the variance of Chevron's return, the residual variance is $1 - 0.7 = 0.3$, so with pay risk proportional to slope times standard deviation the slope can rise by

$$\frac{1}{\sqrt{0.3}} = 1.826 \tag{17}$$

for the same risk borne by the CEO.

Step 2 — Chevron: free cash flow. A mature, cash-rich oil company is prone to overinvest (G&T p. 637: oil firms kept exploring in the early 1980s despite falling prices). The contract should not reward size or reserves added; it should charge for capital employed (economic profit, EVA-type measures) and reward payouts, possibly together with debt that drains free cash.

Step 3 — Chrysler: more CEO-specific value, more distress risk. An automaker's value depends more on the CEO's own strategic choices (product design, cost cutting, labour relations), so a higher pay-for-performance sensitivity on the stock price is worth having. The industry cycle is also common to GM and Ford and could be filtered by relative performance, but Result 18.10's disadvantage bites here: in a concentrated oligopoly of three domestic producers, pay for beating rivals encourages aggressive market-share wars that lower industry profits. Chevron sells into a world market at a price it does not set, so relative performance there has little competitive side effect. Chrysler's history of financial distress also means a CEO with undiversified wealth may be too conservative; some option convexity helps, but near distress options can encourage asset substitution against creditors.

Step 4 — Summary.

	Chrysler (autos)	Chevron (oil)
Main extraneous risk	industry demand cycle	oil price
Relative/indexed pay	limited (competitive side effects)	strongly indicated
Main agency problem	strategy, cost control; distress	free cash flow, overinvestment
Performance measure	stock-based, high sensitivity	indexed stock; capital charge

Answer . Chevron's CEO should be paid on performance net of the oil price (indexed options or peer-relative bonuses) with measures that charge for capital, to curb overinvestment of free cash flow. Chrysler's CEO should have a high stock-based sensitivity because value depends more on his

decisions, with less reliance on explicit relative performance (it would fuel aggressive competition among the three automakers) and attention to risk-taking incentives near distress.

Intuition. Good contracts pay for what the CEO controls and target each firm's own agency problem: luck and free cash flow at Chevron, strategy and distress at Chrysler.

Common mistakes.

- Giving both CEOs the same plain stock options.
- Recommending relative performance everywhere without its competitive side effect.

Book answer. No answer in the book.

Ref: G&T §18.3, pp. 637–638; §18.5, pp. 643–650, Results 18.9–18.11.

8. High Tech: new equity or a partially owned subsidiary? — Grinblatt & Titman 18.3, p. 653

Statement.

Source: Grinblatt, Ch. 18, Exercise 18.3, p. 653 (PDF 666) · Answer in book: no

18.3. John Jacobs, the CEO of High Tech Industries, owns 51 percent of the shares of his $50 million company. The firm is starting a new project that requires $25 million in new equity capital. Jacobs is considering two ways to fund the project. The first is to issue $25 million in new equity. The second is to form a partially owned subsidiary of High Tech, which would be called Super Tech, and have the subsidiary issue the equity. Under the second proposal, Super Tech would be 55 percent owned by High Tech and 45 percent owned by new shareholders. Describe how the incentives of the managers of the new business and John Jacobs are likely to be affected by the two proposals.

What is asked. Jacobs owns 51% of High Tech ($50m). A $25m project can be financed (1) by a High Tech equity issue or (2) through a subsidiary, Super Tech, 55% owned by High Tech and 45% by new investors. How do the two choices affect the incentives of Jacobs and of the new business's managers?

Approach. Jensen–Meckling (an owner-manager with a smaller stake consumes more perks; Amaro Prop. 3.1.2), control and the market for corporate control (G&T §18.1), and the use of a subsidiary's own stock to pay divisional managers (Result 18.12, carve-outs and partial takeovers).

Notation.

- α_1: Jacobs's share of High Tech after option (1); α_2^{ST}: his look-through share of Super Tech under (2).
- Assumption: shares are sold at fair prices and the project has zero NPV, so High Tech is worth $75m after option (1).

Step 1 — Option (1): Jacobs's stake. His shares are worth $0.51 \times 50 = 25.5$. After the issue,

$$\alpha_1 = \frac{25.5}{50 + 25} = \frac{(25.5)}{(75)} = 0.34. \tag{18}$$

He loses his majority: he can now be outvoted, removed or taken over. His cash-flow stake in everything High Tech does falls from 51% to 34%, so each $1 of perks or slack costs him 34 cents instead of 51 (Jensen–Meckling: perks rise, value falls).

Step 2 — Option (1): the new business's managers. If paid in High Tech stock, their project is only $25/75 = 1/3$ of the firm, and the stock price mixes their performance with the existing business. The link between their effort and their pay is weak (Result 18.11).

Step 3 — Option (2): Jacobs's stakes. He keeps 51% of High Tech and therefore control and his incentives in the existing business. His look-through share of the new business is

$$\alpha_2^{ST} = (0.51)(0.55) = 0.2805, \tag{19}$$

lower than the 34% of option (1).

Step 4 — Option (2): the new business's managers. Super Tech has its own traded stock. Its managers can be paid in Super Tech shares or options whose price reflects their own decisions, and the market price monitors the unit (Fisher-Price after its spin-off from Quaker Oats, p. 651; partial takeovers, Result 18.12).

Step 5 — Option (2): a new conflict. Jacobs controls both firms but owns different shares of each. Shifting \$1 from Super Tech to High Tech (transfer prices, allocation of opportunities, overhead charges) changes his wealth by

$$0.51 - 0.2805 = 0.2295 > 0, \tag{20}$$

so he gains 23 cents per dollar taken from Super Tech's minority. New investors anticipate this and pay less. (At a fair price, \$25m for 45% values Super Tech at $25/0.45 = 55.6$ post-money, so High Tech's contribution must be worth $0.55 \times 55.6 = 30.6$.)

Answer . Option (1) dilutes Jacobs to 34% (losing control and weakening his incentives) and gives the new unit's managers only High Tech stock, a weak measure of their own performance. Option (2) keeps Jacobs at 51% of High Tech and lets Super Tech's managers be paid in their own unit's stock (stronger incentives and market monitoring), but cuts his look-through share of the new business to 28% and gives him an incentive (23 cents per dollar) to shift value from Super Tech to High Tech at the expense of its minority shareholders.

Intuition. A subsidiary with its own stock creates a performance measure for the unit; the price is a controlling owner whose cash-flow rights differ across the two firms.

Common mistakes.

- Ignoring the control consequence: 34% is no longer a majority.
- Forgetting the parent–subsidiary conflict that the second structure creates.

Book answer. No answer in the book.

Ref: G&T §18.1, pp. 629–630; §18.5 "Compensation Issues, Mergers, and Divestitures", pp. 650–651, Results 18.11–18.12; Amaro de Matos §3.1.1, pp. 64–66.

9. Cybertex's leveraged recapitalization — Grinblatt & Titman 18.8, pp. 653–654

Statement.

Source: Grinblatt, Ch. 18, Exercise 18.8, pp. 653–654 (PDF 666) · Answer in book: no

18.8. Cybertex's management currently owns 1 percent of the firm's outstanding shares. The firm is currently financed with 50 percent debt and 50 percent equity but is planning to increase its leverage ratio to 80 percent debt by borrowing and using the funds to repurchase shares. Management has decided not to participate in the repurchase so their percentage ownership of the firm will increase.

Explain how managers' investment incentives are likely to change after the recapitalization. Specifically, discuss their incentives to take:

- Negative *NPV* projects that benefit them personally.
- Risky projects.
- Long-term projects that take more than 10 years to provide an adequate return to capital.

What is asked. Leverage rises from 50% to 80% debt through a buyback in which managers do not sell. How do their incentives change toward (i) negative-NPV projects with private benefits, (ii) risky projects, (iii) long-term projects?

Approach. Three tools: ownership and perks (Jensen–Meckling), debt as discipline (Result 18.7) and equity as a call option on the firm (risk shifting, Amaro §3.1.3), and debt overhang plus distress fear for long horizons (G&T Ch. 16; managers' fear of bankruptcy, §18.3).

Notation.

- Firm value normalized to $V = 100$, unchanged by the recap; buyback at a fair price.
- $E_0 = 50$, $E_1 = 20$: equity before and after; managers' stake value $0.01 \times 50 = 0.5$.

Step 1 — Managers' ownership after the recap. The firm borrows $80 - 50 = 30$ and buys back 30 of equity; managers keep their shares, so

$$\frac{0.5}{E_1} = \frac{(0.5)}{(20)} = 0.025. \tag{21}$$

Ownership rises from 1% to 2.5%, a 2.5-fold increase in skin in the game.

Step 2 — (i) Negative-NPV projects with private benefits. Each \$1 wasted now costs managers 2.5 cents instead of 1, so by (7) fewer such projects pass. In addition, interest and principal drain free cash flow and the bankruptcy threat disciplines (Result 18.7). Both effects reduce value-destroying investment.

Step 3 — (ii) Risky projects. Two opposing forces. (a) Equity is now a more levered claim, a call option on firm value with a higher strike: risk raises its value at the creditors' expense, and managers own more of it, so the asset-substitution incentive grows. (b) Managers' human capital and undiversified wealth are tied to the firm, and with 80% debt the probability of distress, and of losing their jobs (only 43% of CEOs keep them, Gilson), is much higher. If most of their wealth is human capital, (b) dominates and they become more conservative; if their pay is option-heavy, (a) dominates.

Step 4 — (iii) Long-term projects. Fewer. Fixed debt payments leave less internal cash; with high leverage part of a project's value goes to creditors (debt overhang); and managers who fear distress before a 10-year payoff arrives prefer projects that pay off quickly (Result 18.4). The larger ownership stake partly offsets this, but only to the extent the long-run value accrues to equity.

Answer . Ownership rises from 1% to 2.5%. (i) Fewer negative-NPV pet projects: managers bear 2.5 times more of the cost and debt service removes free cash. (ii) Ambiguous: levered equity rewards risk, but higher distress risk and undiversified human capital make managers more conservative; the latter likely dominates unless pay is option-based. (iii) Fewer long-term projects: debt overhang, cash commitments and fear of distress shorten managers' horizons.

Intuition. A recap concentrates ownership and removes slack; it fixes the overinvestment problem and creates short-horizon and risk-shifting problems in exchange.

Common mistakes.

- Stating that the recap only adds risk-taking: managers' own distress costs push the other way.
- Forgetting that the ownership increase comes from shrinking equity, not from managers buying.

Book answer. No answer in the book.

Ref: G&T §18.3, pp. 637–638 (Result 18.4); §18.4, pp. 639–641 (Results 18.6–18.7); Amaro de Matos §3.1.3, p. 71.

10. Stock options in the Jensen–Meckling model — Amaro de Matos 3.1.5 #3, p. 77

Statement.

Source: Amaro de Matos, Ch. 3, Exercise 3.1.5 #3, p. 77 (PDF 95) · Answer in book: yes, p. 256

3. Some authors argue that one way to control the conflicts of interests between managers and stockholders is to give stock options to managers. How can we explain this argument using the simple model of section 3.1.1? What other factors should be included in this analysis?

Context needed to solve it *From Amaro de Matos §3.1.1 "Agency Costs of Outside Equity", pp. 64–66 (PDF 82–84):*

Consider a firm where the manager owns a fraction of the equity. This manager-owner will make operating decisions maximizing the joint effect of two different sources of utility. The manager's utility will be assumed to be generated not only by the benefits derived from pecuniary returns, but also by some nonpecuniary aspects of the manager's activity. Examples of nonpecuniary aspects are the physical appointments of the office, the kind and amount of contributions to charity, attractiveness of the staff, nature of personal relationship with employees, and an excellent computer.

When the owner-manager sells equity to outsiders, a typical agency problem arises because of the difference of interests between him/her and the outside shareholders.

Consider a firm generating a flow with present market value $\overline{V}$, and let F denote the market value of the stream of the manager's expenditures on nonpecuniary benefits. This reduces the effective value of the firm to $V = \overline{V} - F$. If the owner-manager holds a fraction α of the shares of the firm, its utility will be described by the real function

$$U(\alpha V, F),$$

which is increasing and concave in both arguments. In this context, two main results are in order. First, given an optimal choice of F utility to the owner-manager decreases as new equity is issued. Second, the effective value of the firm decreases. In fact, the decline in the value of the firm is entirely imposed on the owner-manager. This is reflected in the decrease of his/her utility.

For fixed α, the optimal value $F^*(\alpha)$ is obtained maximizing the utility

$$\max_F U[\alpha(\overline{V} - F), F].$$

In other words, $F^*(\alpha)$ must satisfy the first-order condition

$$-\alpha U_1 + U_2 = 0. \qquad (3.1)$$

Let $\xi(\alpha)$ denote the optimal value of the utility, or

$$\xi(\alpha) \equiv U[\alpha(\overline{V} - F^*(\alpha)), F^*(\alpha)].$$

Using the first-order condition (3.1), it is clear that $\xi(\alpha)$ is a monotonic increasing function of α, since

$$d\xi(\alpha) = (-\alpha U_1 + U_2)\, dF^* + V\, d\alpha = V\, d\alpha.$$

This means that as α increases, the optimal utility of the owner-manager increases. Issuing equity is just the reverse process, reducing the value of α. Thus, the first result comes:

Proposition 3.1.1. As new equity is issued, the utility to the owner-manager decreases.

It was just stressed that the decrease in the utility comes from the fact that the owner-manager carries all the reduction in value of the firm when new equity is issued. Indeed, since

$$V = \overline{V} - F,$$

it follows that $dV = -dF$. In order to show that the value increases with α, it suffices to show that F decreases. This follows since $dF/d\alpha$ can be obtained differentiating the first-order condition (3.1) and using the envelope theorem. It follows that

$$\frac{d^2U}{dF^2}\frac{dF}{d\alpha} = U_1.$$

By assumption $U_1 > 0$ and since these derivatives are calculated at the optimum,

$$\frac{d^2U}{dF^2} < 0,$$

leading to

$$\frac{dF}{d\alpha} < 0.$$

Thus, this confirms the second result:

Proposition 3.1.2. As new equity is issued, the effective value of the firm decreases.

It is clear from the preceding formalism that the reason why the value of the firm decreases is that the optimal market value F^* of the nonpecuniary benefits increases. The managers would have an incentive to

convince the outsiders that the value of the firm will remain at the level $V^* = \overline{V} - F^*$ and the fringe benefits will be kept at F^*. Jensen and Meckling (1976) point out that this suggestion is not credible if the outsiders are rational. Basically, they know that it would be optimal for the management to stay at V^* only if they hold the fraction α of the shares.

Under the assumption that $\overline{V}$ is fixed, the new shareholders know beforehand that managers will be tempted to increase their benefits, given that now they only bear a smaller fraction of the costs associated with F. That is why selling some shares will decrease the value of the firm to a level $V^\circ < V^*$, with benefits at a level $F^\circ > F^*$. Knowing that, the new shareholders in fact would not be willing to pay more than the fraction of the new value V° corresponding to their fraction of shares.

In order to solve this problem, the management may accept being controlled by the outsiders. This can be done either by systematic auditing (to see whether $F = F^*$), by formal control systems, by budget restrictions, by incentive compensations to the management, or by the market for corporate control.

From Amaro de Matos §3.1.2, "Executive Stock Options", pp. 70–71 (PDF 88–89):

An interesting alternative to help solving the principal–agent problem in the context of management's compensation is to offer a package including call options on shares of the firm. Such options are typically offered *at the money*—that is to say, at an exercise price coinciding with the current value of the shares (say, S). Recalling that the payoff of such options at time T is

$$\max[0, S_T - S],$$

this component in an executive compensation package will give the right incentive for the manager to care about S_T, the future value of the shares.

Among other advantages, it is clear that the introduction of such instruments as a payment device align the incentives of managers with those of the shareholders. In fact, the residual claim to shareholders, as expressed in (2.2), can be seen as an option on the value of the firm with an exercise price that coincides with the face value of debt. If the management is given options, that constrains the real value of the payment to the performance of the firm, as reflected in the market value of the stock.

A second advantage is that this type of instrument allows the reduction of the base pay of the management, thus reducing salary differences among different employees.

Finally, in this setting, the payment of the manager is at risk. At least, this component of the package may be worthless at time T, which justifies a real commitment with the goals of the firm.

What is asked. Use the §3.1.1 owner-manager model to explain why stock options can reduce manager–shareholder conflicts, and list what else the analysis should include.

Approach. In §3.1.1 the manager chooses perks F trading off his share α of firm value against private benefits; Proposition 3.1.2 shows that F falls (value rises) when α rises. An option grant raises the manager's marginal share of value, which acts like a higher α.

Notation.

- $\overline{V}$: firm value without perks; F: market value of perks; $V = \overline{V} - F$: effective value.
- $\alpha \in (0, 1]$: manager's ownership share; $U(\alpha V, F)$: his utility, increasing and concave, U_1, U_2 partial derivatives.
- n: options granted as a fraction of shares; $\Delta \in (0, 1)$: option delta (change in option value per unit change in share value).

Step 1 — The manager's problem and first-order condition. He solves $\max_F U[\alpha(\overline{V} - F), F]$. Differentiating with respect to F:

$$g(F, \alpha) \equiv \frac{\partial}{\partial F} U[\alpha(\overline{V} - F), F] = U_1 \cdot (-\alpha) + U_2 = 0, \tag{22}$$

which is (3.1). The second-order condition is $g_F = \alpha^2 U_{11} - 2\alpha U_{12} + U_{22} < 0$.

Step 2 — How perks respond to α. Differentiate (22) with respect to α at $F^*(\alpha)$:

$$\begin{aligned} g_F \frac{dF^*}{d\alpha} + g_\alpha &= 0, \\ g_\alpha &= -U_1 - \alpha U_{11}(\overline{V} - F) + U_{21}(\overline{V} - F), \\ \frac{dF^*}{d\alpha} &= \frac{U_1 - (\overline{V} - F)(U_{12} - \alpha U_{11})}{g_F}. \end{aligned} \tag{23}$$

Without wealth effects ($U_{11} = U_{12} = 0$, e.g. $U = \alpha V + h(F)$ with h concave) this reduces to $g_F\, dF^*/d\alpha = U_1$, the book's expression, and since $U_1 > 0$ and $g_F < 0$,

$$\frac{dF^*}{d\alpha} = \frac{U_1}{g_F} < 0, \qquad \frac{dV}{d\alpha} = -\frac{dF^*}{d\alpha} > 0. \tag{24}$$

(With strong wealth effects the second term in (23) could in principle offset U_1; the book's argument neglects it.)

Step 3 — Options act like a higher α. The manager's pecuniary claim becomes $\alpha V + n \cdot$ (option payoff). Near the current price an option's value moves by Δ per unit of share value, so his marginal share of V is

$$\alpha_{\text{eff}} = \alpha + n\Delta > \alpha, \tag{25}$$

and by (24) perks fall and value rises. This is the book's statement that giving options "is equivalent to an increase in α".

Step 4 — Illustration (assumed functional form). Let $U = \alpha(\overline{V} - F) + 2c\sqrt{F}$ with $c = 0.5$, $\overline{V} = 100$. The first-order condition $-\alpha + c/\sqrt{F} = 0$ gives

$$F^*(\alpha) = \left(\frac{c}{\alpha}\right)^2. \tag{26}$$

At $\alpha = 0.10$: $F^* = (0.5/0.10)^2 = 25$, $V = 100 - 25 = 75$. Grant at-the-money options on $n = 0.10$ of the shares with $\Delta = 0.5$:

$$\begin{aligned} \alpha_{\text{eff}} &= (0.10) + (0.10)(0.5) = 0.15, \\ F^* &= \left(\frac{0.5}{0.15}\right)^2 = 11.11, \qquad V = 100 - 11.11 = 88.89, \end{aligned} \tag{27}$$

a gain of $25 - 11.11 = 13.89$. Here $dF^*/d\alpha = 1/h''(F) < 0$ with $h''(F) = -\frac{c}{2}F^{-3/2}$.

Step 5 — What else must enter the analysis.

- *Convexity and risk*: option value rises with volatility. This counters managers' excessive risk aversion (and under-leverage) but can induce excessive risk and wealth transfers from bondholders (asset substitution, §3.1.3).
- *Cost and dilution*: options are paid by existing shareholders; exercise issues new shares.
- *Delta is not constant*: deep out of the money, $\Delta \to 0$ and the incentive vanishes (pressure to reprice).
- *Manager risk aversion*: an undiversified manager values options below their cost; the stock moves for reasons outside his control (G&T Result 18.9), so indexing helps.
- *Short-termism and manipulation*: pay tied to the price at a date invites managing that price (Tirole §7.2.1).
- *Effort*: §3.1.1 fixes $\overline{V}$; with an effort choice, options also raise $\overline{V}$.

Answer . In §3.1.1 perks fall and value rises with the manager's share α ($dF/d\alpha = U_1/g_F < 0$). Stock options raise his marginal share of value to $\alpha + n\Delta$, so they act like an increase in α and reduce perk consumption (e.g. V from 75 to 88.9). The analysis should add the risk-taking effect of convex payoffs (including wealth transfers from bondholders), dilution and cost, the instability of delta, the manager's risk aversion and the noise in stock prices, and manipulation of short-term prices.

Intuition. Options make the manager a larger residual claimant at the margin without requiring him to buy shares.

Common mistakes.

- Treating an option like a share: its delta is below 1 and changes with the price.
- Ignoring that the extra incentive is paid for by existing shareholders and creditors.

Book answer. Matches the book (p. 256): giving options "is equivalent to an increase in α, leading to an increase in the firm's value"; the book lists the same other factors (risk-taking and underleverage, dilution on exercise, transfer from bondholders). Note the text's $d\xi = V\,d\alpha$ should read $U_1 V\,d\alpha$; the sign is unaffected.

Ref: Amaro de Matos §3.1.1, pp. 64–66; §3.1.2 "Executive Stock Options", pp. 70–71; §3.1.3, p. 71.

11. The Pareto-efficient fee schedule — Amaro de Matos 3.1.5 #4, p. 77

Statement.

Source: Amaro de Matos, Ch. 3, Exercise 3.1.5 #4, p. 77 (PDF 95) · Answer in book: yes, pp. 256–257

4. Explain condition (3.4) and show what happens to the optimal value f*[w(a°, θ)] when θ changes.

Context needed to solve it *From Amaro de Matos §3.1.2 "Principal–Agent Problems", "Determining Optimal Fee Schedules", pp. 66–68 (PDF 84–86):*

The first formalization of the problem of determining an optimal fee schedule was made by Ross (1973). In this model the management takes an *unobservable* action that determines the probability distribution of the shareholders' payoff. The payment to the management is a function of the payoff, which is observable. The fee schedule is determined optimally by the shareholders in order to provide the incentives for the managers to choose the best possible actions when maximizing their utility. This maximization is constrained to providing managers a reservation expected utility.

Notice that if the actions of the management *are observable* and managers are risk averse, a first-best solution is attainable by making the fee very low, unless the optimal action (from the shareholders' viewpoint) is taken by the management. In other words, the payment schedule will share the risk associated with the managers' action optimally. If the actions are not observable, as in Ross's model, the expected utility for the shareholders will be lowered by their ignorance, since they will have to find a *costly* way to induce the managers to take an optimal action. The difference in the expected utility to the shareholders reflects the agency cost of monitoring the management through the payment schedule.

The problem is to define an optimal fee schedule f to the management such that it will be in the interest of the managers to take the actions desired by the shareholders.

Let a be the action chosen by the management, defined in a space $\mathcal{A}$ of feasible actions. Uncertainty is described by the set of states of nature Ω together with a probability distribution over the elements of Ω. As before, for $w \in \Omega$, the expectation of a random variable $x(w) \in R^{\Omega}$ on Ω is denoted by $E(x)$.

Let the result of action a be a payoff $\theta(a, w)$ in the state of nature $w \in \Omega$, which is assumed to be *unknown* beforehand to both management and shareholders. This payoff is thus a real valued function $\theta(a, w)$: $\mathcal{A} \times \Omega \to R$.

The fee f is assumed here to be a function *uniquely* of the payoff, that is,

$$f \equiv f(\theta(a, w)).$$

Managers and shareholders are supposed to have their preferences represented respectively by the utility functions G and U. The management faces the problem of choosing an optimal action a° as a function of a jointly optimal fee schedule f^* to be determined, such that

$$\max_{a\in\mathcal{A}} E\{G[f^*(\theta(a,w))]\}, \qquad (3.2)$$

constrained to a minimum level of expected utility k

$$E\{G[f^*(\theta(a^\circ,w))]\} > k.$$

Hence, the optimal action taken by the managers depends on the established fee schedule f^*. It will be denoted by $a(f^*)$. On the other hand, the shareholders will have to choose the optimal fee schedule f^* so it will be the schedule f satisfying the following variational optimization:

$$\max_{f} E\{U[\theta(a(f),w) - f(\theta(a(f),w))]\} \qquad (3.3)$$

subject to

$$E\{G[f(\theta(a^\circ,w))]\} > k.$$

The problem can be considerably simplified if one looks for a specific type of solution. Here, two kinds of fee schedules are proposed, following Ross's model.

1. A family of fee schedules can be achieved if managers and shareholders cooperate, maximizing a weighted sum of their expected utilities. In this case, f satisfies

   $$\max_{f} E\{U[\theta - f] + \lambda G[f]\}.$$

   where λ parameterizes the family, defining the weights of both utilities. This family of fees is said to be *Pareto-efficient*. When U and G are both monotonic and concave, the optimal fee schedule must maximize the argument of the expected value and therefore is defined implicitly by the equation

   $$U'[\theta - f] = \lambda G'[f]. \qquad (3.4)$$

2. An alternative approach to find a fee schedule is to assume that, at optimality, the utilities of managers and shareholders are positively and linearly related. That is to say, the optimal fee schedule must satisfy the *similarity condition*

   $$U[\theta - f] = cG[f] + b, \qquad (3.5)$$

   for constants c and b such that $c > 0$. It is clear that both managers and shareholders have the same attitude toward risky payoffs, and hence this similarity condition ensures that the managers will always choose the act more desired by the shareholders.

Ross argues that (3.4) and (3.5) are simultaneously satisfied only if both utility functions U and G are in the linear risk tolerance class. This will imply that the optimal fee schedule f is a *linear* function of the payoff.

What is asked. Interpret condition (3.4), $U'[\theta - f] = \lambda G'[f]$, and find how the optimal fee f^* changes across states, i.e. when the payoff changes.

Approach. (3.4) is the first-order condition of maximizing a weighted sum of the two parties' expected utilities state by state: Pareto-efficient (first-best) risk sharing, valid when the manager's action is observable. Differentiating it across states gives the sharing slope in terms of absolute risk aversions.

Notation.(The exercise writes the payoff as $w(a^\circ,\theta)$ with state θ; the text writes $\theta(a,w)$ with state w. We follow the exercise.)

- w: firm payoff in state θ, $w_\theta \equiv dw/d\theta$; $f(w)$: manager's fee, $f_\theta \equiv df/d\theta$.
- U: shareholders' (principal's) utility of $w - f$; G: manager's (agent's) utility of f; both increasing and concave.
- $\lambda > 0$: weight on the manager's utility; $R^P = -U''/U'$, $R^A = -G''/G'$: absolute risk aversions.

Step 1 — Where (3.4) comes from. Pareto efficiency maximizes $E\{U[w-f]+\lambda G[f]\}$. Since f can be chosen separately in each state, maximize the integrand pointwise:

$$\frac{\partial}{\partial f}\left(U[w-f]+\lambda G[f]\right) = -U'[w-f]+\lambda G'[f] = 0 \implies U'[w-f] = \lambda G'[f], \tag{28}$$

with second-order condition $U''+\lambda G'' < 0$, true by concavity.

Step 2 — Interpretation. Writing (28) for two states w and w' and dividing,

$$\frac{U'[w-f(w)]}{U'[w'-f(w')]} = \frac{G'[f(w)]}{G'[f(w')]}\,: \tag{29}$$

both parties have the same marginal rate of substitution between income in any two states, so no reallocation of risk can make one better off without hurting the other. $\lambda = U'/G'$ is the same in every state; its value picks the point on the Pareto frontier (it is pinned down by the manager's reservation utility k). This is first best: it presumes the desired action is enforced directly.

Step 3 — Differentiate across states. Differentiate (28) with respect to θ:

$$U''[w-f]\,(w_\theta - f_\theta) = \lambda G'''[f]\,f_\theta.$$

Substitute $\lambda = U'/G'$ from (28):

$$U''\,(w_\theta - f_\theta) = \frac{U'}{G'}\,G''\,f_\theta.$$

Divide both sides by U':

$$\frac{U''}{U'}(w_\theta - f_\theta) = \frac{G''}{G'}f_\theta \iff -R^P(w_\theta - f_\theta) = -R^A f_\theta.$$

Collect the f_θ terms:

$$\begin{aligned} R^P w_\theta &= (R^P + R^A)\,f_\theta, \\ f_\theta &= \frac{R^P}{R^P + R^A}\,w_\theta. \end{aligned} \tag{30}$$

Step 4 — Reading the slope. The fee moves in the same direction as the payoff, absorbing a share $R^P/(R^P+R^A) \in [0,1]$ (equivalently $\tau^A/(\tau^A+\tau^P)$ with risk tolerances $\tau = 1/R$) of every change. Example: $R^P = 1$, $R^A = 3$ gives a share of $1/(1+3) = 0.25$.

- Risk-neutral shareholders ($R^P = 0$): $f_\theta = 0$, a constant fee; shareholders bear all risk.
- Risk-neutral manager ($R^A = 0$): $f_\theta = w_\theta$; the manager bears all risk and shareholders get a fixed rent.
- Constant R^P, R^A (CARA): the slope is constant and f is linear (Exercise 12).

The check script also verifies (30) numerically for $U = -e^{-x}$ and $G = \ln f$, where $R^A = 1/f$ varies.

Answer . (3.4) is Pareto-efficient risk sharing: equal marginal rates of substitution across states, $\lambda = U'/G'$ constant. Across states the fee changes by $df/d\theta = \dfrac{R^P}{R^P+R^A}\,dw/d\theta$: a constant fee if shareholders are risk neutral, the whole payoff (minus a fixed rent) if the manager is risk neutral.

Intuition. Risk is shared in proportion to risk tolerance: the less risk-averse party absorbs more of each shock.

Common mistakes.

- Reading (3.4) as an incentive condition: it is pure risk sharing with observable actions.
- Inverting the ratio: the manager's share uses the *principal's* risk aversion in the numerator.

Book answer. Matches the book (pp. 256–257): $df/d\theta = \dfrac{R_a^P}{R_a^P + R_a^A}\dfrac{dw}{d\theta}$, with the same two limiting cases (Shavell 1979).

Ref: Amaro de Matos §3.1.2 "Determining Optimal Fee Schedules", pp. 66–68.

12. CARA utilities and a linear fee schedule — Amaro de Matos 3.1.5 #6, p. 77

Statement.

Source: Amaro de Matos, Ch. 3, Exercise 3.1.5 #6, p. 77 (PDF 95) · Answer in book: yes, p. 257

6. Prove that when both utility functions V and G have constant absolute risk averse coefficients, the optimal fee schedule $f(\cdot)$ is a linear function of the payoff.

Context needed to solve it *From Amaro de Matos §3.1.2, "Determining Optimal Fee Schedules", pp. 67–68 (PDF 85–86):*

Let a be the action chosen by the management, defined in a space $\mathcal{A}$ of feasible actions. [...] Let the result of action a be a payoff $\theta(a, w)$ in the state of nature $w \in \Omega$, which is assumed to be *unknown* beforehand to both management and shareholders. [...]

The fee f is assumed here to be a function *uniquely* of the payoff, that is,

$$f \equiv f(\theta(a, w)).$$

Managers and shareholders are supposed to have their preferences represented respectively by the utility functions G and U. [...]

1. A family of fee schedules can be achieved if managers and shareholders cooperate, maximizing a weighted sum of their expected utilities. In this case, f satisfies

$$\max_f E\{U[\theta - f] + \lambda G[f]\}.$$

where λ parameterizes the family, defining the weights of both utilities. This family of fees is said to be *Pareto-efficient*. When U and G are both monotonic and concave, the optimal fee schedule must maximize the argument of the expected value and therefore is defined implicitly by the equation

$$U'[\theta - f] = \lambda G'[f]. \qquad (3.4)$$

2. An alternative approach to find a fee schedule is to assume that, at optimality, the utilities of managers and shareholders are positively and linearly related. That is to say, the optimal fee schedule must satisfy the *similarity condition*

$$U[\theta - f] = cG[f] + b, \qquad (3.5)$$

for constants c and b such that $c > 0$. [...]

Ross argues that (3.4) and (3.5) are simultaneously satisfied only if both utility functions U and G are in the linear risk tolerance class. This will imply that the optimal fee schedule f is a *linear* function of the payoff.

What is asked. Prove that if both parties have constant absolute risk aversion, the optimal (Pareto-efficient) fee schedule is linear in the payoff. (The statement's "V" is the shareholders' utility U of the text.)

Approach. Apply (3.4) with exponential utilities; the exponentials turn the condition into a linear equation in f after taking logs. Consistent with Exercise 11: constant R^P, R^A make the slope constant.

Notation.

- $U(x) = -\frac{1}{\delta}e^{-\delta x}$ (shareholders, $x = w - f$), $G(f) = -\frac{1}{\alpha}e^{-\alpha f}$ (manager); $\delta, \alpha > 0$.
- $\lambda > 0$: Pareto weight; w: payoff.

Step 1 — Check the risk aversions. $U'(x) = e^{-\delta x}$, $U''(x) = -\delta e^{-\delta x}$, so $-U''/U' = \delta$; likewise $G'(f) = e^{-\alpha f}$, $-G''/G' = \alpha$. Both constant.

Step 2 — Write (3.4).

$$U'[w - f] = \lambda G'[f] \iff e^{-\delta(w-f)} = \lambda e^{-\alpha f}. \qquad (31)$$

Step 3 — Take logs and solve for f.

$$\begin{aligned} -\delta(w-f) &= \ln\lambda - \alpha f, \\ -\delta w + \delta f + \alpha f &= \ln\lambda, \\ (\alpha+\delta)f &= \delta w + \ln\lambda, \\ f(w) &= \frac{\delta}{\alpha+\delta}\,w + \frac{\ln\lambda}{\alpha+\delta}. \end{aligned} \tag{32}$$

This is linear in w: the slope $\delta/(\alpha+\delta) = R^P/(R^P+R^A)$ is Exercise 11's share, now constant; λ only moves the intercept (it is set by the manager's reservation utility k).

Step 4 — Similarity also holds. Substitute (32): $w - f = \frac{\alpha}{\alpha+\delta}w - \frac{\ln\lambda}{\alpha+\delta}$, so

$$U(w-f) = -\tfrac{1}{\delta}\exp\Big(-\tfrac{\alpha\delta}{\alpha+\delta}w + \tfrac{\delta\ln\lambda}{\alpha+\delta}\Big), \qquad G(f) = -\tfrac{1}{\alpha}\exp\Big(-\tfrac{\alpha\delta}{\alpha+\delta}w - \tfrac{\alpha\ln\lambda}{\alpha+\delta}\Big),$$

and the ratio is

$$\frac{U(w-f)}{G(f)} = \frac{\alpha}{\delta}\exp\Big(\tfrac{(\delta+\alpha)\ln\lambda}{\alpha+\delta}\Big) = \frac{\alpha\lambda}{\delta}, \tag{33}$$

a positive constant: (3.5) holds with $c = \alpha\lambda/\delta > 0$, $b = 0$. Pareto efficiency, similarity and linearity hold together, as Ross's result says (CARA is in the linear-risk-tolerance class).

Step 5 — Numbers (illustration). $\delta = 2$, $\alpha = 6$: slope $2/8 = 0.25$; with $\lambda = 1$ the intercept is 0, with $\lambda = e$ it is $1/8 = 0.125$. The script checks (31) and (33) at several w.

Answer . With $U = -\frac{1}{\delta}e^{-\delta(w-f)}$ and $G = -\frac{1}{\alpha}e^{-\alpha f}$, condition (3.4) gives $f = \dfrac{\delta}{\alpha+\delta}w + \dfrac{\ln\lambda}{\alpha+\delta}$, linear in the payoff with slope $R^P/(R^P+R^A)$; it also satisfies similarity with $c = \alpha\lambda/\delta$.

Intuition. With constant risk aversion the efficient share of each extra dollar does not depend on how rich either party already is, so the same share applies at every payoff.

Common mistakes.

- Swapping δ and α in the slope.
- Claiming this proves linear contracts are optimal under hidden action: it concerns the Pareto-efficient (first-best) schedule (3.4).

Book answer. Matches the book (p. 257): $f = \dfrac{\delta}{\alpha+\delta}w + \dfrac{\ln\lambda}{\alpha+\delta}$.

Ref: Amaro de Matos §3.1.2, pp. 67–68, conditions (3.4)–(3.5).

13. Congruence and the power of the incentive scheme — Tirole 3.10, p. 148

Statement.

Source: Tirole, Ch. 3, Exercise 3.10, p. 148 (PDF 161) · Answer in book: yes, p. 575

Exercise 3.10 (congruence, pledgeable income, and power of incentive scheme). The credit rationing model developed in this chapter assumes that the entrepreneur's and investors' interests are *a priori* dissonant, and that incentives must be aligned by giving the entrepreneur enough of a stake in the case of success.

Suppose that the entrepreneur and the investors have indeed dissonant preferences with probability x, but have naturally aligned interests with probability $1-x$. Which prevails is unknown to both sides at the financing stage and is discovered (only) by the entrepreneur just before the moral-hazard stage.

More precisely, consider the fixed-investment model of Section 3.2. The investors's outlay is $I - A$ and they demand an expected rate of return equal to 0. The entrepreneur is risk neutral and protected by limited liability. With probability x, interests are dissonant: the entrepreneur obtains private benefit B by misbehaving (the probability of success is p_L) and 0 by behaving (probability of success p_H). With probability $1-x$, interests are aligned: the entrepreneur's taking her private benefit B coincides with choosing probability of success p_H.

(i) Consider a "simple incentive scheme" in which the entrepreneur receives R_b in the case of success and 0 in the case of failure. R_b thus measures the "power of the incentive scheme."

Show that it may be optimal to choose a low-powered incentive scheme if preferences are rather congruent (x low) and that the incentive scheme is necessarily high-powered if preferences are rather dissonant (x high).

(ii) Show that one cannot improve on simple incentive schemes by presenting the entrepreneur with a menu of two options (two outcome-contingent incentive schemes) from which she will choose once she learns whether preferences are congruent or dissonant.

Context needed to solve it *From Tirole §3.2.1 "The Fixed-Investment Model", pp. 115–116 (PDF 128–129):*

Variants of the following entrepreneurial model will be used in the following: an entrepreneur (also called the "insider" or the "borrower," "she") has a project. This project requires a fixed investment I. The entrepreneur initially has "assets" ("cash on hand" or "net worth") $A < I$. For the moment we interpret these assets as being cash or liquid securities that can be used toward covering the cost of investment. [...] The entrepreneur's cash can either be invested in the project or used for consumption. To implement the project the entrepreneur must borrow $I - A$ from lenders. [...]

Project. If undertaken, the project either succeeds, that is, yields verifiable income $R > 0$, or fails, that is, yields no income. The probability of success is denoted by p. The project is subject to moral hazard. The entrepreneur can "behave" ("work," "exert effort," "take no private benefit") or "misbehave" ("shirk," "take a private benefit") [...].

Behaving yields probability $p = p_H$ of success and no private benefit to the entrepreneur, and misbehaving results in probability $p = p_L < p_H$ of success and private benefit $B > 0$ (measured in units of account) to the entrepreneur. In the "effort interpretation," B can also be interpreted as a disutility of effort saved by the entrepreneur when shirking. Let $\Delta p \equiv p_H - p_L$.

Preferences and the loan agreement. Both the borrower and the potential lenders (or "investors") are risk neutral. For notational simplicity, there is no time preference; the rate of return expected by investors (which is also the riskless rate, due to risk neutrality) is taken to be 0. The borrower is protected by limited liability, and so her income cannot take negative values.

Lenders behave competitively in the sense that the loan, if any, makes zero profit. [...]

Let us turn to the loan contract. A contract first stipulates whether the project is financed. If so, it further specifies how the profit is shared between the lenders and the borrower. The borrower's limited liability will imply that both sides receive 0 in the case of failure [...]. In the case of success, the two parties share the profit R; R_b goes to the borrower and R_l to the lenders. To sum up, we posit an incentive scheme for the entrepreneur of the following form: R_b in the case of success, 0 in the case of failure.

[...] We summarize the timing in Figure 3.1.

Figure 3.1 (timeline, left to right): Loan agreement (sharing rule) → Investment → Moral hazard → Outcome.

We assume that the project is viable only in the absence of moral hazard. That is, the project has positive NPV if the entrepreneur behaves,

$$p_H R - I > 0, \qquad (3.1)$$

but negative NPV, even if one includes the borrower's private benefit, if she does not,

$$p_L R - I + B < 0. \qquad (3.2)$$

From Tirole §3.2.2 "The Lenders' Credit Analysis", pp. 116–117 (PDF 129–130):

[...] the borrower will therefore behave if the following "incentive compatibility constraint" is satisfied:

$$p_H R_b \geqslant p_L R_b + B \quad \text{or} \quad (\Delta p) R_b \geqslant B. \qquad (\text{IC}_b)$$

From this incentive compatibility constraint we infer that the highest income in the case of success that can be pledged to the lenders without jeopardizing the borrower's incentives is

$$R - \frac{B}{\Delta p}.$$

The *(expected) pledgeable income* is then

$$\mathcal{P} = p_H \left(R - \frac{B}{\Delta p} \right).$$

Because the lenders must break even in order to be willing to finance the project, a necessary condition for the borrower to receive a loan is that the expected pledgeable income exceed the lenders' initial outlay:

$$\mathcal{P} \equiv p_H \left(R - \frac{B}{\Delta p} \right) \geqslant I - A, \qquad (\text{IR}_l)$$

[...] Thus a necessary condition for financing to be arranged is

$$A \geqslant \bar{A} = p_H \frac{B}{\Delta p} - (p_H R - I). \qquad (3.3)$$

[...] Conversely, if $A \geqslant \bar{A}$, the entrepreneur is able to secure financing, and so condition (3.3) is both a necessary and a sufficient condition for financing. The entrepreneur offers claim R_l to competitive investors so as not to leave them with a surplus:

$$p_H R_l = I - A.$$

[...] Using the breakeven condition for the lenders ($p_H R_l = I - A$), the borrower obtains net utility or payoff (where "net" means that we subtract the consumption utility, A, that the entrepreneur would get by not undertaking the project):

$$U_b = \begin{cases} 0 & \text{if } A < \bar{A}, \\ p_H R_b - A = p_H(R - R_l) - A = p_H R - I & \text{if } A \geqslant \bar{A}. \end{cases} \qquad (3.6)$$

As could have been expected from the zero-profit condition for the lenders, *the borrower receives the entire social surplus or net present value if the project is funded.*

What is asked. (i) With probability x the entrepreneur's interests conflict with the investors' (misbehaving gives B and p_L); with probability $1 - x$ they are aligned (she gets B and success probability p_H anyway). Show that a low-powered scheme (R_b below $B/\Delta p$) can be optimal when x is low and that the scheme must be high-powered when x is high. (ii) Show that a menu of two schemes, chosen after she learns the state, cannot do better than a simple scheme.

Approach. Tirole's fixed-investment model (§3.2): the incentive constraint (IC_b) $\Delta p R_b \geq B$, pledgeable income and the investors' break-even. A high-powered scheme buys good behaviour in the dissonant state at the cost of a rent $p_H B/\Delta p$; a low-powered one keeps the whole of R pledgeable but accepts misbehaviour when interests diverge.

Notation.

- I, A, R, p_H, p_L, B, Δp as in §3.2; $x \in (0,1)$: probability of dissonance.
- R_b: entrepreneur's reward on success (power of the scheme); 0 on failure.
- $\mathcal{P}$: expected pledgeable income; NPV^1, NPV^2: NPV under the high- and low-powered scheme.

Step 1 — High-powered scheme, $R_b \geq B/\Delta p$. She behaves in both states (in the congruent state she also collects B). NPV and pledgeable income (with the smallest incentive-compatible $R_b = B/\Delta p$):

$$\text{NPV}^1 = p_H R - I + (1 - x)B, \qquad (34)$$

$$\mathcal{P}^1 = p_H\Big(R - \frac{B}{\Delta p}\Big) \geq I - A. \qquad (35)$$

Step 2 — Low-powered scheme, $R_b < B/\Delta p$. In the dissonant state she misbehaves (p_L, B); in the congruent state she behaves and gets B. $R_b = 0$ maximizes pledgeable income:

$$\text{NPV}^2 = x(p_L R + B) + (1 - x)(p_H R + B) - I, \qquad (36)$$

$$\mathcal{P}^2 = [x p_L + (1 - x)p_H]R = p_H R - x\,\Delta p R \geq I - A. \qquad (37)$$

Step 3 — Compare NPVs. Subtract (36) from (34):

$$\begin{aligned}\text{NPV}^1 - \text{NPV}^2 &= p_H R + (1-x)B - xp_L R - xB - (1-x)p_H R - (1-x)B \\ &= xp_H R - xp_L R - xB = x(\Delta pR - B) > 0, \end{aligned} \tag{38}$$

since (3.1) and (3.2) give $p_H R - I > 0 > p_L R - I + B$, hence $\Delta pR > B$. The entrepreneur, who receives the NPV, prefers the high-powered scheme whenever it can be financed.

Step 4 — Compare pledgeable incomes. From (35)–(37),

$$\mathcal{P}^2 > \mathcal{P}^1 \iff p_H R - x\Delta pR > p_H R - \frac{p_H B}{\Delta p} \iff x < \hat{x} \equiv \frac{p_H B}{(\Delta p)^2 R}. \tag{39}$$

Step 5 — Optimal power. If (35) holds: high-powered. If it fails but (37) holds (possible only for $x < \hat{x}$): low-powered, sacrificing $x(\Delta pR - B)$ of NPV to obtain funding. If $x \geq \hat{x}$, $\mathcal{P}^2 \leq \mathcal{P}^1$: whenever the project is financed at all, (35) holds and the scheme is high-powered.

Step 6 — Illustration (parameter point). $p_H = 0.9$, $p_L = 0.5$, $R = 10$, $B = 0.8$, $I = 8$: (3.1) $9 - 8 > 0$, (3.2) $5 - 8 + 0.8 < 0$; $B/\Delta p = 2$, $\mathcal{P}^1 = 0.9(10-2) = 7.2$, $\hat{x} = 0.72/1.6 = 0.45$.

- $x = 0.2$, $A = 0.4$ ($I - A = 7.6$): $\mathcal{P}^1 = 7.2 < 7.6$ but $\mathcal{P}^2 = 9 - 0.8 = 8.2 \geq 7.6$: low-powered, $\text{NPV}^2 = 0.2(5.8) + 0.8(9.8) - 8 = 1.0$ versus the unattainable $\text{NPV}^1 = 1.64$; the gap is $0.2(4 - 0.8) = 0.64$.
- $x = 0.6$, $A = 1$ ($I - A = 7$): $\mathcal{P}^2 = 9 - 2.4 = 6.6 < 7 \leq 7.2$: high-powered.

Step 7 — (ii) Menus do not help. Let the menu offer (R_b^S, R_b^F) for the dissonant state and $(\hat{R}_b^S, \hat{R}_b^F)$ for the congruent one, all ≥ 0; neither state is observed by investors, so she picks the scheme she likes. NPV depends only on behaviour, so compare pledgeable incomes for each behaviour pattern.

- *Dissonant type behaves.* Her scheme must satisfy $\Delta p(R_b^S - R_b^F) \geq B$, so her expected pay is $p_H R_b^S + (1-p_H)R_b^F \geq R_b^F + p_H B/\Delta p \geq p_H B/\Delta p$. The congruent type behaves under any scheme and must not prefer the dissonant one: $p_H \hat{R}_b^S + (1-p_H)\hat{R}_b^F \geq p_H R_b^S + (1-p_H)R_b^F \geq p_H B/\Delta p$. Expected pay is at least $p_H B/\Delta p$ in both states, so pledgeable income is at most $p_H(R - B/\Delta p) = \mathcal{P}^1$: the simple high-powered scheme (take $\hat{R} = R$, $R_b^S = B/\Delta p$, $R_b^F = 0$) attains it.
- *Dissonant type misbehaves.* Pay is non-negative, so pledgeable income is at most $[xp_L + (1-x)p_H]R = \mathcal{P}^2$, attained by the simple scheme $R_b = 0$.

For example, a congruent-state scheme $(1, 0)$ beside a dissonant-state $(2, 0)$ fails: the congruent type would take $(2, 0)$, worth $0.9 \times 2 = 1.8 > 0.9$.

Answer (i). High-powered ($R_b = B/\Delta p$) maximizes NPV, $\text{NPV}^1 - \text{NPV}^2 = x(\Delta pR - B) > 0$, and is used whenever $p_H(R - B/\Delta p) \geq I - A$. Low-powered ($R_b = 0$) raises pledgeable income to $p_H R - x\Delta pR$, above $p_H(R - B/\Delta p)$ iff $x < \hat{x} = p_H B/[(\Delta p)^2 R]$, so it is optimal when x is low and the high-powered scheme cannot be financed. For $x \geq \hat{x}$ any financed scheme is high-powered.

Answer (ii). The congruent type can always take the dissonant-state scheme, so a menu cannot lower the rent below $p_H B/\Delta p$ (when the dissonant type behaves) or below 0 (when she does not); simple schemes already achieve these bounds.

Intuition. Incentive pay is costly only when interests diverge. When they rarely do, it can pay to forgo incentives, accept occasional misbehaviour and pledge the whole return; when they often do, incentives are indispensable.

Common mistakes.

- Forgetting the private benefit B in the congruent state when computing NPV.
- Comparing NPVs only: the choice is driven by pledgeable income whenever the high-powered scheme cannot be financed.

- In (ii), ignoring the congruent type's option to take the dissonant-state scheme.

Book answer. Matches the book (p. 575): $\text{NPV}^1 = p_H R - I + (1-x)B$ with financing condition $p_H(R - B/\Delta p) \geq I - A$; $\text{NPV}^2 < \text{NPV}^1$ with condition $[xp_L + (1-x)p_H]R \geq I - A$; low-powered only if x is low; in (ii) the congruent type must not pretend, so one may as well take $\hat{R} = R$.

Ref: Tirole §3.2.1–3.2.2, pp. 115–117.

14. Incentives versus efficient risk sharing — Amaro de Matos 3.1.5 #5, p. 77

Statement.

Source: Amaro de Matos, Ch. 3, Exercise 3.1.5 #5, p. 77 (PDF 95) · Answer in book: yes, p. 257

5. Show that the risk-sharing rule implied by condition U′/G′ = λ − ψ d(w_a/w_b)/dθ is not Pareto-efficient when the agent is risk averse. Is it when the agent is risk neutral?

Context needed to solve it *From Amaro de Matos §3.1.2 "Principal–Agent Problems", "Determining Optimal Fee Schedules", pp. 66–70 (PDF 84–88):*

The first formalization of the problem of determining an optimal fee schedule was made by Ross (1973). In this model the management takes an *unobservable* action that determines the probability distribution of the shareholders' payoff. The payment to the management is a function of the payoff, which is observable. The fee schedule is determined optimally by the shareholders in order to provide the incentives for the managers to choose the best possible actions when maximizing their utility. This maximization is constrained to providing managers a reservation expected utility.

Notice that if the actions of the management *are observable* and managers are risk averse, a first-best solution is attainable by making the fee very low, unless the optimal action (from the shareholders' viewpoint) is taken by the management. In other words, the payment schedule will share the risk associated with the managers' action optimally. If the actions are not observable, as in Ross's model, the expected utility for the shareholders will be lowered by their ignorance, since they will have to find a *costly* way to induce the managers to take an optimal action. The difference in the expected utility to the shareholders reflects the agency cost of monitoring the management through the payment schedule.

[...] Let a be the action chosen by the management, defined in a space $\mathcal{A}$ of feasible actions. Uncertainty is described by the set of states of nature Ω together with a probability distribution over the elements of Ω. [...] Let the result of action a be a payoff $\theta(a, w)$ in the state of nature $w \in \Omega$, which is assumed to be *unknown* beforehand to both management and shareholders. This payoff is thus a real valued function $\theta(a, w)$: $\mathcal{A} \times \Omega \to R$.

The fee f is assumed here to be a function *uniquely* of the payoff, that is,

$$f \equiv f(\theta(a, w)).$$

Managers and shareholders are supposed to have their preferences represented respectively by the utility functions G and U. The management faces the problem of choosing an optimal action a° as a function of a jointly optimal fee schedule f^* to be determined, such that

$$\max_{a \in \mathcal{A}} E\{G[f^*(\theta(a, w))]\}, \qquad (3.2)$$

constrained to a minimum level of expected utility k

$$E\{G[f^*(\theta(a^\circ, w))]\} > k.$$

Hence, the optimal action taken by the managers depends on the established fee schedule f^*. It will be denoted by $a(f^*)$. On the other hand, the shareholders will have to choose the optimal fee schedule f^* so it will be the schedule f satisfying the following variational optimization:

$$\max_{f} E\{U[\theta(a(f), w) - f(\theta(a(f), w))]\} \qquad (3.3)$$

subject to

$$E\{G[f(\theta(a^\circ, w))]\} > k.$$

[...]

1. A family of fee schedules can be achieved if managers and shareholders cooperate, maximizing a weighted sum of their expected utilities. In this case, f satisfies

$$\max_f E\{U[\theta - f] + \lambda G[f]\}.$$

where λ parameterizes the family, defining the weights of both utilities. This family of fees is said to be *Pareto-efficient*. When U and G are both monotonic and concave, the optimal fee schedule must maximize the argument of the expected value and therefore is defined implicitly by the equation

$$U'[\theta - f] = \lambda G'[f]. \qquad (3.4)$$

[...]

In the original problem, there is the restriction that the expected utility of the managers should not be below a certain level. Besides, there is a constraint that the managers will act so as to optimize their own expected utility, which is expressed by the first-order condition

$$E\{G'[f(\theta)]f'(\theta)\theta_a\} = 0, \qquad (3.6)$$

where θ_a denotes the derivative of θ with respect to a and $k = 0$. Then, the shareholders' problem is to maximize with respect to f the Lagrangean

$$\mathcal{L} = E\{U[\theta - f(\theta)] + \psi G'[f(\theta)]f'(\theta)\theta_a + \lambda G[f(\theta)]\},$$

where ψ and λ are the Lagrange multipliers associated respectively with the restrictions expressed in (3.6) and (3.3). Under some general conditions the Euler-Lagrange equation characterizes optimality provided that

$$\frac{U'}{G'} = \lambda - \psi \frac{d}{d\theta}\left(\frac{\theta_a}{\theta_w}\right).$$

Comparing this solution with (3.4), it follows that in general the fee schedule will be Pareto efficient if either

1. $\psi = 0$, meaning that (3.6) is not binding and the fee schedules will be Pareto-efficient independently of the payoff function θ, or
2. The term $d(\theta_a/\theta_w)/dw$ is a function only of a.

For the optimal fee schedule to be Pareto efficient independently of the payoff function $\theta(a, w)$, the first of these two conditions has to be fulfilled, implying a linear fee schedule, as shown by Ross, and therefore linear risk tolerance utility functions U and G.

[...]

Most of the subsequent literature in this topic is centered on the case where the managers are averse to effort and the shareholders try to maximize the value of the firm (i.e., are risk neutral). For instance, Shavell (1979) shows that under these conditions a Pareto optimal fee schedule does not maximize the expected profits of the shareholders. This, once again, reflects the costs of monitoring the actions of the management.

In principle, the risk associated with the output is shared among the management and the stockholders. The way this is done strongly depends on the risk-aversion characteristics of both shareholders and managers, as discussed by Shavell. Even in the case of risk neutrality of the shareholders, the fee schedule cannot be fixed as a constant value because then the managers would have no incentive to overcome their aversion to effort. The payment schedule has to depend on the output, leading as before to a second-best solution.

It may be proved that the shareholders will use any indication of the level of effort of the management in the equilibrium fee schedule, in order to approach its first-best solution. Information, even if incomplete, thus has value. It is also clear that the less reliable the proxy for the managers' effort is, the worse off the shareholders will be (i.e., the more distant their solution is from the first-best).

In conclusion, the perfect knowledge of the action would destroy the agency costs associated with the monitoring of the management.

What is asked. Show that the second-best rule $U'/G' = \lambda - \psi \frac{d}{d\theta}(\theta_a/\theta_w)$ is not Pareto-efficient when the manager is risk averse, and say whether it is when he is risk neutral. (The statement's w_a/w_b is the text's θ_a/θ_w.)

Approach. Compare with (3.4): Pareto efficiency requires U'/G' constant across states. When the incentive constraint binds ($\psi > 0$) the extra term varies across states, so the ratio does too. Following

the Shavell paragraph of the text, the manager is averse to effort; with risk-neutral shareholders Pareto efficiency means a constant fee, which kills incentives. A two-outcome version makes every step explicit.

Notation.

- Shareholders risk neutral ($U' = 1$); manager's utility $G(f) - c(a)$ with $G' > 0$, $G'' < 0$, effort cost $c'(a) > 0$.
- λ, ψ: multipliers on the participation and incentive constraints.
- Two-outcome version: payoff θ_H or θ_L; effort raises $\Pr(\theta_H)$ from p_L to p_H; fees f_H, f_L; reservation utility k.

Step 1 — Pareto efficiency means full insurance. With $U' = 1$, (3.4) reads $1 = \lambda G'(f)$ in every state, so $G'(f)$ and hence f are constant: the manager is fully insured.

Step 2 — A constant fee gives no incentive. With effort cost, the manager's incentive constraint (the analogue of (3.6)) is $E\{G'(f)f'(\theta)\theta_a\} = c'(a) > 0$. A constant fee has $f' = 0$, the left side is 0, and he chooses the least effort. To implement more effort the fee must vary with the payoff ($f' \neq 0$), so $G'(f)$ varies across states, $U'/G' = 1/G'$ is not constant and (3.4) fails: the multiplier ψ on the incentive constraint is positive and the rule is second best.

Step 3 — Two-outcome example: the second-best contract. Take $G = \sqrt{f}$, $c = 1$, $k = 4$, $p_H = 0.8$, $p_L = 0.4$ (illustrative). Write $u_s = \sqrt{f_s}$. The binding constraints are

$$\text{(IC)} \quad (p_H - p_L)(u_H - u_L) = c \;\Rightarrow\; u_H - u_L = \frac{(1)}{(0.4)} = 2.5, \tag{40}$$

$$\text{(IR)} \quad p_H u_H + (1 - p_H)u_L - c = k \;\Rightarrow\; u_L + p_H(2.5) = 5 \;\Rightarrow\; u_L = 5 - (0.8)(2.5) = 3, \tag{41}$$

so $u_H = 5.5$, $f_L = 3^2 = 9$, $f_H = 5.5^2 = 30.25$, expected pay $0.8(30.25) + 0.2(9) = 26$. With observable effort the first best pays a constant $f = (k + c)^2 = 25$; the agency cost is $26 - 25 = 1$.

Step 4 — The second-best rule and its multipliers. The shareholders maximize $\sum_s p_H^s(\theta_s - f_s)$ subject to (IR) with multiplier λ and (IC) with multiplier ψ. The first-order condition in f_s is

$$-p_H^s + \lambda p_H^s G'(f_s) + \psi(p_H^s - p_L^s)G'(f_s) = 0 \implies \frac{1}{G'(f_s)} = \lambda + \psi\,\frac{p_H^s - p_L^s}{p_H^s}, \tag{42}$$

the discrete analogue of $U'/G' = \lambda - \psi\, d(\theta_a/\theta_w)/d\theta$: the likelihood-ratio term plays the role of $d(\theta_a/\theta_w)/d\theta$. Here $1/G'(f) = 2\sqrt{f}$, so $1/G'(f_H) = 2(5.5) = 11$, $1/G'(f_L) = 2(3) = 6$; the likelihood terms are $(0.8 - 0.4)/0.8 = 0.5$ and $(0.2 - 0.6)/0.2 = -2$. Solving $\lambda + 0.5\psi = 11$, $\lambda - 2\psi = 6$:

$$2.5\psi = 5 \;\Rightarrow\; \psi = 2, \qquad \lambda = 11 - 0.5(2) = 10. \tag{43}$$

$\psi > 0$ and U'/G' takes the values 11 and 6: the marginal rates of substitution differ across states, so the sharing rule is not Pareto-efficient. (Check: $\lambda = E_H[1/G'] = 0.8(11) + 0.2(6) = 10$.)

Step 5 — Risk-neutral manager. With $G(f) = f$, (3.4) gives $U'(\theta - f) = \lambda$, i.e. $\theta - f$ constant: sell the firm to the manager, $f = \theta - K$. He is the residual claimant, so his privately optimal effort is the efficient one; the incentive constraint does not bind ($\psi = 0$) and the rule *is* Pareto-efficient (and first best). With $\theta_H = 40$, $\theta_L = 10$: expected output $0.8(40) + 0.2(10) = 34$, $K = 34 - c - k = 29$, $f_H = 11$, $f_L = -19$; he exerts effort since $0.4(40 - 10) = 12 \geq 1$; shareholders get 29, the first-best surplus. This requires the manager to absorb losses ($f_L < 0$); with limited liability (Tirole §3.2) a risk-neutral agent still earns a rent and pledgeable income falls short of NPV.

Answer . With a risk-averse manager, incentives require the fee to vary with the payoff, so U'/G' varies across states ($\psi > 0$; e.g. 11 vs 6, $\psi = 2$, $\lambda = 10$), violating (3.4): risk sharing is sacrificed for incentives, at an agency cost (1 in the example). With a risk-neutral manager the optimum sells him the firm ($f = \theta - K$), $\psi = 0$ and the rule is Pareto-efficient.

Intuition. Insurance and incentives pull in opposite directions when the agent is risk averse; when he is risk neutral, making him residual claimant delivers both.

Common mistakes.

- Forgetting that without effort aversion (as in the text's $G[f]$) there is nothing to motivate.
- Reading $\psi = 0$ as "no incentive problem" in general: it holds when the agent is risk neutral and unconstrained in wealth.

Book answer. Matches the book (p. 257): with a risk-neutral principal, Pareto efficiency means full insurance, a fixed fee gives zero effort, so the fee must depend on the payoff and the rule is not an optimal risk sharing; with a risk-neutral agent the optimum gives him the payoff minus a fixed income to the principal (Shavell 1979).

Ref: Amaro de Matos §3.1.2, pp. 66–70, (3.4), (3.6) and the Euler–Lagrange condition.

15. Optimality of golden parachutes — Tirole 7.9, p. 326

Statement.

Source: Tirole, Ch. 7, Exercise 7.9, p. 326 (PDF 339) · Answer in book: yes, pp. 604–605

Exercise 7.9 (optimality of golden parachutes). Return to the manipulation model of Section 7.2.1, with the possibility of informed manipulation. Confirm the heuristic analysis of that section through a careful analysis, allowing for general contracts (the reward R_b^r or R_b^q is contingent on the revealed information and may *a priori* exceed $B/\Delta p$; a fixed payment can be made in both states and only under revealed poor prospects: L_l^r and $L_l^q \leqslant L$; allow $q_H R$ to be larger or smaller than L).

Context needed to solve it *From Tirole §3.2.1–3.2.2 (the fixed-investment model), pp. 115–117 (PDF 128–130):*

[...] an entrepreneur [...] has a project. This project requires a fixed investment I. The entrepreneur initially has "assets" ("cash on hand" or "net worth") $A < I$. [...] To implement the project the entrepreneur must borrow $I - A$ from lenders. [...] If undertaken, the project either succeeds, that is, yields verifiable income $R > 0$, or fails, that is, yields no income. [...] Behaving yields probability $p = p_H$ of success and no private benefit to the entrepreneur, and misbehaving results in probability $p = p_L < p_H$ of success and private benefit $B > 0$ (measured in units of account) to the entrepreneur. [...] Let $\Delta p \equiv p_H - p_L$. [...] Both the borrower and the potential lenders (or "investors") are risk neutral. [...] the rate of return expected by investors (which is also the riskless rate, due to risk neutrality) is taken to be 0. The borrower is protected by limited liability, and so her income cannot take negative values. Lenders behave competitively in the sense that the loan, if any, makes zero profit. [...]

[...] the borrower will therefore behave if the following "incentive compatibility constraint" is satisfied:

$$p_H R_b \geqslant p_L R_b + B \quad \text{or} \quad (\Delta p) R_b \geqslant B. \qquad (\mathrm{IC}_b)$$

From Tirole §7.2.1.1 "Managerial Myopia: The Incentive for Posturing", pp. 301–303 (PDF 314–316):

Consider the fixed-investment model of Section 3.2, with the new ingredient that there is some learning at an intermediate stage about the entrepreneur's ability to run the project and the concomitant opportunity to replace her (or to liquidate the project) on the basis of this information (see Figure 7.7). The manager's type, which is a synonym for the probability of success, is either $r_\bullet$ or $q_\bullet$, where the dot subscript refers to the fact that the probability of success is not solely determined by the manager's ability and is a function of later effort (high or low).

Figure 7.7 (timeline, left to right): Funding stage. Entrepreneur invests I, of which $I - A$ is borrowed from investors. → Manipulation? → Probability of success ($r_\bullet$ or $q_\bullet$) learned publicly. → Liquidation/replacement (yields L), otherwise continue → Moral hazard (high or low effort). → Outcome (success: R, failure: 0).

No manipulation. Let us for the moment rule out any managerial manipulation of the intermediate information received by the investors. The latter learn at the intermediate stage that the probability of success in the case of continuation $\pi_\bullet = (\pi_H, \pi_L)$ (that is, contingent on effort: π_H in the case of good behavior, π_L in the case of misbehavior) is either high ($r_\bullet = (r_H, r_L)$) or low ($q_\bullet = (q_H, q_L)$), with $r_H > q_H$ and $r_L > q_L$. At the funding stage, no one knows which prevails, and the prior on the two possibilities is $(\alpha, 1-\alpha)$:

$$\pi_\bullet = \begin{cases} r_\bullet & \text{with probability } \alpha, \\ q_\bullet & \text{with probability } 1-\alpha. \end{cases}$$

[...] For simplicity, we will also assume that the manager's "type" ($r_\bullet$ or $q_\bullet$) is orthogonal to later moral hazard. So, if r_H and q_H denote the probabilities of success in the case of good behavior, and r_L and q_L those in the case of misbehavior, then

$$r_H - r_L = q_H - q_L = p_H - p_L = \Delta p,$$

letting $p_H \equiv \alpha r_H + (1-\alpha) q_H$ and $p_L \equiv \alpha r_L + (1-\alpha) q_L$ refer to the prior beliefs that the project will succeed under good and bad behavior, respectively. Thus, regardless of the manager's type, shirking reduces the probability of success by Δp.

Suppose that, when it accrues, the information about the manager's type is public and, for simplicity, verifiable; and that, contingent on the signal, the initial contract specifies whether management is allowed to continue or not. In the case of termination, the firm generates an expected profit L that can be shared between investors and incumbent management.

Example. Suppose that the manager's replacement is another similar manager of unknown ability in the job. Then, the "liquidation" value is

$$L = p_H \left(R - \frac{B}{\Delta p} \right).$$

For, the new management must be provided with a reward, $B/\Delta p$, in the case of success that induces the high effort. The pledgeable income is therefore equal to $p_H[R - B/\Delta p]$.

We make the following assumption:

$$q_H R > L. \qquad (7.9)$$

Inequality (7.9) says that, ceteris paribus, even a low-ability manager would prefer to keep her job, as this yields a higher NPV than termination. Put differently, it is *ex ante* efficient for the entrepreneur to retain her job. [...]

We will also assume that

$$I - A > \max \left\{ p_H \left(R - \frac{B}{\Delta p} \right), L \right\}, \qquad (7.10)$$

while

$$I - A < \alpha r_H \left(R - \frac{B}{\Delta p} \right) + (1-\alpha) L. \qquad (7.11)$$

The first inequality (7.10) states that guarantee of either tenure (the entrepreneur always keeps her job) or termination (the entrepreneur is always fired) does not generate enough pledgeable income to attract investors. Because the incentive problem is independent of the entrepreneur's type, the entrepreneur must be rewarded at least $B/\Delta p$ for success in order to have an incentive to behave. And so the pledgeable income under guaranteed tenure is insufficient to cover the investors' initial outlay, $I - A$. By contrast, the second inequality, (7.11), which requires that

$$q_H \left(R - \frac{B}{\Delta p} \right) < L < r_H \left(R - \frac{B}{\Delta p} \right),$$

implies that there is enough pledgeable income to attract investors when there is termination in case of low ability, provided that the investors receive the return L in the case of termination.

Under a competitive capital market the entrepreneur's utility in case of funding is equal to the NPV:

$$U_b(z^r, z^q) = \alpha[z^r(r_H R) + (1 - z^r)L] + (1-\alpha)[z^q(q_H R) + (1 - z^q)L] - I,$$

where z^r and z^q are the contracted-for probabilities of continuation of employment of a high- and low-ability entrepreneur, respectively. From (7.9), this utility is maximized by a guaranteed tenure:

$$z^r = z^q = 1.$$

Guaranteed tenure, however, does not attract funds (from (7.10)); and so some (contingent) termination must be conceded in order to satisfy the investors' breakeven constraint:

$$\alpha \left[z^r r_H \left(R - \frac{B}{\Delta p} \right) + (1 - z^r) L_l^r \right] + (1-\alpha) \left[z^q q_H \left(R - \frac{B}{\Delta p} \right) + (1 - z^q) L_l^q \right] \geqslant I - A,$$

where L_l^r and L_l^q ($\leqslant L$) are the lenders' returns in the case of termination of a high- and low-ability manager, respectively. Clearly, setting

$$L_l^r = L_l^q = L$$

is optimal since this relaxes the investors' breakeven constraint without altering the NPV. Also, it is more efficient (in terms of maximizing both the NPV and the pledgeable income) to retain a high-ability manager:

$$z^r = 1.$$

Let $z^q = z^*$. From (7.10) and (7.11), the value $z^* \in (0,1)$ is the smallest value that satisfies the investors' breakeven constraint:

$$\alpha r_H \left(R - \frac{B}{\Delta p}\right) + (1-\alpha)\left[z^* q_H \left(R - \frac{B}{\Delta p}\right) + (1-z^*)L\right] = I - A. \qquad (7.12)$$

Some termination in case of low ability is the concession made by the entrepreneur to attract investors. As is familiar, the entrepreneur sacrifices value (NPV) to boost pledgeable income.

Manipulation. Until now, we have assumed that the information received by the investors at the intermediate stage lies outside the entrepreneur's control. Let us now assume that the entrepreneur can, at a cost, alter this information.

More precisely, suppose that the entrepreneur can generate the high signal $r_\bullet$ by (secretly) manipulating the information. This manipulation comes at a cost: the probability of success falls (uniformly) by $\tau > 0$.

We distinguish two forms of manipulation:

Uninformed manipulation. The entrepreneur does not know her type when deciding whether to manipulate information (so, she learns her type at the same time as the investors in Figure 7.7).

Informed manipulation. The entrepreneur knows her type when choosing whether to manipulate the information (but learns it only after the funding stage, which therefore still occurs under symmetric information).

Suppose, in the first step, that the financing contract specifies, besides the probabilities of continuation z^r and z^q for signals $r_\bullet$ and $q_\bullet$, a reward $R_b \geqslant B/\Delta p$ in the case of continuation and success (and no payment to the entrepreneur otherwise). [...]

From Tirole §7.2.1.1, pp. 303–304 (PDF 316–317):

Under *informed manipulation*, the entrepreneur is tempted to manipulate the information only when she learns that she is inefficient. The new no-manipulation constraint is

$$z^r[(q_H - \tau)R_b] \leqslant z^q[q_H R_b]$$

or

$$\frac{z^r}{z^q} \leqslant \frac{1}{1 - \tau/q_H}. \qquad (7.14)$$

[...] The relevant no-manipulation constraint, (7.13) in the case of uninformed manipulation and (7.14) in the case of informed manipulation, will be labeled (NM).

Whether manipulation is uninformed or informed, the initial contract must in general lower z^r or increase z^q—in a nutshell, make continuation less signal dependent—or both. [...]

The threat of manipulation may prevent the firm from receiving funding in the first place. Start from the solution ($z^r = 1$, $z^q = z^*$) as defined by equation (7.12) when there is no scope for manipulation, and suppose that the ratio $1/z^*$ does not satisfy the relevant (NM) constraint. That is, [...]

$$\frac{1}{z^*} > \frac{1}{1 - \tau/q_H}$$

under informed manipulation.

If one keeps $z^r \equiv 1$, then z^q must be increased above z^* so as to satisfy the (NM) constraint. Increasing z^q above z^*, however, is not feasible since this reduces pledgeable income, which then becomes insufficient to cover the investors' initial outlay. Thus continuation cannot be guaranteed to a high-ability entrepreneur ($z^r < 1$).

This reduction in z^r reduces pledgeable income as $r_H[R - B/\Delta p] > L$. Hence, z^q must also be brought down below z^* in order to make up for the shortfall in pledgeable income.

As one could have expected, *the entrepreneur's ability to cook the books ex post may jeopardize funding ex ante.* And, even if funding is feasible, this ability reduces the NPV.

From Tirole §7.2.1.2 "Golden Parachutes", pp. 304–305 (PDF 317–318):

Top managers often receive very large compensation packages when their employment is terminated. These "golden parachutes" appear particularly "obscene" when termination is motivated by poor performance. Of course, many of these packages result from the board being in cahoots (or not wanting to enter any conflict) with top management. There is also some efficiency rationale for golden parachutes. Intuitively, the "softened landing" that they offer to managers makes them less prone to engage in various venal behaviors, such as earnings manipulations, in order to keep their job. In a nutshell, proponents of golden parachutes argue that they are the price to pay for incentive compatibility.

Are golden parachutes beneficial here? They are clearly costly as they reduce pledgeable income. However, a golden parachute helps relax the (NM) constraint. In a sense, they create more "balanced" incentives for the manager by increasing her payoff in the case of liquidation. Indeed, consider (for example) the case of informed manipulation and suppose that the entrepreneur receives some amount $T \geqslant 0$ when admitting that prospects are poor, i.e., when the signal is $q_\bullet$. The new (NM) constraint is

$$z^r[(q_H - \tau)R_b] \leqslant z^q[q_H R_b] + T, \qquad (7.15)$$

where $R_b = B/\Delta p$ in order to maximize the income that can be pledged to investors.

The key question is whether it is cheaper to prevent manipulation by making tenure relatively insensitive to new information or by granting a golden parachute ($T > 0$). To answer this question, let us write the NPV (which does not depend on T),

$$U_b(z^r, z^q, T) = \text{NPV} = \alpha[L + z^r(r_H R - L)] + (1-\alpha)[L + z^q(q_H R - L)] - I,$$

and the pledgeable income,

$$\mathcal{P}(z^r, z^q, T) = \alpha[L + z^r[r_H(R - R_b) - L]] + (1-\alpha)[L + z^q[q_H(R - R_b) - L] - T] = I - A.$$

Intuitively, there are two "currencies" available for paying the manager: continuation and golden parachute. A golden parachute is just a cash transfer while the continuation policy affects the NPV. One would therefore expect a golden parachute to be used exactly when the continuation policy is an inefficient policy, that is, when continuation under poor prospects reduces the NPV.

To demonstrate this "efficient currency result," suppose that $T > 0$. Looking at the pledgeable income, a unit increase in z^q (which increases the *ex ante* utility U_b because of the assumption that $q_H R > L$, but reduces pledgeable income), must be compensated by a decrease in the golden parachute equal in absolute value to

$$\left|\frac{dT}{dz^q}\right| = L - q_H(R - R_b).$$

From (7.15), this marginal change that keeps investor income constant relaxes the (NM) constraint:

$$dz^q(q_H R_b) + dT = dz^q[q_H R_b - L + q_H(R - R_b)] = (q_H R - L)\, dz^q > 0.$$

Thus the optimal golden parachute policy is to have none:

$$T^* = 0.$$

Not so when continuation under poor prospects reduces the NPV, and not only the investor income. Suppose now that

$$q_H R < L.$$

Then, from the previous reasoning, a golden parachute is a cheaper instrument than an insensitive tenure to keep the entrepreneur from manipulating accounts. When it is optimal to fire the manager in the case of poor prospects, she is paid a golden parachute:

$$z^{q*} = 0 \quad \text{and} \quad T^* = z^r(q_H - \tau)R_b > 0.$$

Exercise 7.9 asks the reader to check this heuristic reasoning more formally.

From Tirole §7.2.1.2, footnotes 51–53, p. 305 (PDF 318):

51. This compensation is slightly different from a golden parachute, since the latter would be received contingent on termination (in particular, when $z^r < 1$, the golden parachute would be received with positive probability even though the manager does not admit to poor prospects). The form of golden parachute considered here is more efficient because it is more effective at addressing the (NM) constraint. But it relies on our assumption that the state of nature is contractible. The analysis would not change much with the alternative formalization.

52. R_b is assumed to be the same in both states, but this involves no loss of generality.
53. So, in the previous notation, $L_l^r = L$ and $L_l^q = L - T/(1 - z^q)$.

What is asked. Under informed manipulation, allow general contracts (signal-contingent rewards that may exceed $B/\Delta p$, payments on termination in both states, lenders' termination returns $L_l^s \leq L$) and check formally the heuristic of §7.2.1.2: no golden parachute when $q_H R > L$; fire the low type and pay a golden parachute $T = z^r(q_H - \tau)B/\Delta p$ when $q_H R < L$.

Approach. Write the contract, the moral-hazard and no-manipulation constraints and the investors' break-even, and note that NPV does not depend on transfers. The key simplification: all that matters for the no-manipulation constraint and for investors is the low type's *expected rent*, whatever its mix of continuation pay and termination pay. Then compare the two "currencies" (continuation versus cash).

Notation.

- Signals $s \in \{r, q\}$ with prior $(\alpha, 1-\alpha)$; continuation probabilities z^r, z^q; reward on continuation and success R_b^s; payment on termination $T^s \geq 0$, so lenders get $L_l^s = L - T^s \leq L$.
- $b \equiv B/\Delta p$; $X \equiv R - b$; τ: fall in success probability from manipulation; $\theta \equiv (q_H - \tau)/q_H$.
- $\Pi \equiv z^q q_H R_b^q + (1 - z^q)T^q$: the q-type's expected payoff when she reports truthfully.

Step 1 — Constraints and objective. Moral hazard after continuation: $R_b^s \geq b$. A q-type who manipulates obtains signal r, continues with probability z^r and succeeds with probability $q_H - \tau$:

$$(\text{NM}') \quad z^r(q_H - \tau)R_b^r + (1 - z^r)T^r \leq z^q q_H R_b^q + (1 - z^q)T^q = \Pi. \tag{44}$$

Investors' break-even:

$$\alpha[z^r r_H(R - R_b^r) + (1 - z^r)(L - T^r)] + (1 - \alpha)[z^q q_H(R - R_b^q) + (1 - z^q)(L - T^q)] \geq I - A. \tag{45}$$

NPV, which the entrepreneur receives, does not depend on transfers:

$$\text{NPV} = \alpha[L + z^r(r_H R - L)] + (1 - \alpha)[L + z^q(q_H R - L)] - I. \tag{46}$$

Step 2 — The r-state terms. R_b^r and T^r appear with a plus sign on the left of (44) and with a minus sign in (45), and not in (46). Lowering them to their floors relaxes both constraints at no cost:

$$R_b^r = b, \qquad T^r = 0. \tag{47}$$

Step 3 — Rewrite in terms of the rent Π. The q-state part of (45) is $z^q q_H R + (1 - z^q)L - \Pi$. The constraints on Π are (NM′), $\Pi \geq z^r(q_H - \tau)b$, and $R_b^q \geq b$, $T^q \geq 0$, which give $\Pi \geq z^q q_H b$. Investors want the smallest rent:

$$\Pi^* = \max\{z^r(q_H - \tau)b,\ z^q q_H b\}, \tag{48}$$

$$\mathcal{P}(z^r, z^q) = \alpha[L + z^r(r_H X - L)] + (1 - \alpha)[L + z^q(q_H R - L) - \Pi^*]. \tag{49}$$

If $z^q \geq \theta z^r$, $\Pi^* = z^q q_H b$ is delivered with $R_b^q = b$, $T^q = 0$. If $z^q < \theta z^r$, the gap $z^r(q_H - \tau)b - z^q q_H b$ must be paid by $T^q > 0$ (a golden parachute) or $R_b^q > b$; for investors the two cost the same. The program is: maximize (46) subject to $\mathcal{P} \geq I - A$.

Step 4 — Case $q_H R > L$: no golden parachute. In the region $z^q < \theta z^r$, Π^* does not depend on z^q, so

$$\frac{\partial \mathcal{P}}{\partial z^q} = (1 - \alpha)(q_H R - L) > 0, \qquad \frac{\partial \,\text{NPV}}{\partial z^q} = (1 - \alpha)(q_H R - L) > 0 : \tag{50}$$

raising z^q (and cutting T^q) raises both NPV and pledgeable income. Hence the optimum has $z^q \geq \theta z^r$, $T^{q*} = 0$, $R_b^q = b$. This is the book's marginal argument: holding investor income fixed, $dT = [q_H(R - R_b) - L]\, dz^q$, and the net relaxation of (NM) is

$$q_H R_b + \frac{dT}{dz^q} = q_H R_b + q_H(R - R_b) - L = q_H R - L > 0. \tag{51}$$

In the region $z^q \geq \theta z^r$, $\mathcal{P}$ is the no-manipulation pledgeable income, so the program is the §7.2.1.1 program plus (NM): $z^q \geq \theta z^r$. If $\theta \leq z^*$ the solution $(1, z^*)$ of (7.12) already satisfies (NM). If $\theta > z^*$, (NM) binds, $z^q = \theta z^r$, and $\mathcal{P}$ is linear in z^r between $\mathcal{P} = L$ at $z^r = 0$ and $\mathcal{P}(1, \theta) < \mathcal{P}(1, z^*) = I - A$ at $z^r = 1$; since $L < I - A$ by (7.10), no contract is funded — the formal version of "the ability to cook the books ex post may jeopardize funding ex ante".

Numbers (parameter point satisfying (7.9)–(7.11)): $\alpha = 0.5$, $r_H = 0.9$, $q_H = 0.5$, $\Delta p = 0.3$, $B = 1.5$ ($b = 5$, $X = 5$), $R = 10$, $L = 4$, $I - A = 4.1$. Then $q_H R = 5 > L$, $q_H X = 2.5 < 4 < 4.5 = r_H X$, $p_H X = 3.5$, $4.1 > \max\{3.5, 4\}$, $4.1 < 0.5(4.5) + 0.5(4) = 4.25$. From (7.12): $2.25 + 0.5[2.5z^* + 4(1 - z^*)] = 4.1 \Rightarrow z^* = 0.2$. With $\tau = 0.45$, $\theta = 0.1 \leq 0.2$: funded at $(1, 0.2)$, no parachute. With $\tau = 0.1$, $\theta = 0.8$: $\mathcal{P}(1, 0.8) = 2.25 + 0.5[0.8(2.5) + 0.2(4)] = 3.65 < 4.1$ and $L = 4 < 4.1$: not funded; a parachute at $(1, 0.2)$ would give $\mathcal{P} = 2.25 + 0.5[0.2(5) + 0.8(4) - 0.4(5)] = 3.35$, worse. In (51): $2.5 + (2.5 - 4) = 1 = q_H R - L$.

Step 5 — Case $q_H R < L$: fire and pay a golden parachute. Now continuation of the low type destroys value: $\partial \text{NPV}/\partial z^q = (1 - \alpha)(q_H R - L) < 0$, and $\partial \mathcal{P}/\partial z^q$ equals $(1 - \alpha)(q_H R - L) < 0$ below θz^r and $(1 - \alpha)(q_H X - L) < 0$ above it. So $z^{q*} = 0$, and with $z^q = 0$ the whole rent is the termination payment:

$$T^{q*} = \Pi^* = z^r(q_H - \tau)\frac{B}{\Delta p} > 0. \tag{52}$$

Then $\mathcal{P}(z^r) = L + z^r[\alpha(r_H X - L) - (1 - \alpha)(q_H - \tau)b]$ and NPV rises with z^r (since $r_H R > L$): z^r is the largest value in $[0, 1]$ with $\mathcal{P} \geq I - A$.

Numbers: $\alpha = 0.5$, $r_H = 0.95$, $q_H = 0.3$, $\Delta p = 0.2$, $B = 0.6$ ($b = 3$, $X = 7$), $R = 10$, $L = 3.5$, $\tau = 0.1$, $I = 5$, $A = 0.5$. Checks: $q_H R = 3 < L$; $q_H X = 2.1 < 3.5 < 6.65 = r_H X$; $p_H X = 4.375$, $4.5 > \max\{4.375, 3.5\}$, $4.5 < 0.5(6.65) + 0.5(3.5) = 5.075$. With $z^r = 1$:

$$\begin{aligned} T^* &= (1)(0.3 - 0.1)(3) = 0.6, \\ \mathcal{P} &= 3.5 + [0.5(6.65 - 3.5) - 0.5(0.2)(3)] = 3.5 + 1.275 = 4.775 \geq 4.5, \\ \text{NPV} &= 0.5(9.5) + 0.5(3.5) - 5 = 1.5. \end{aligned}$$

Without a parachute, (NM) needs $z^q \geq \theta = 2/3$: $\mathcal{P} = 4.608$ and NPV $= 1.333$, lower by $(1 - \alpha)\theta(L - q_H R) = 0.5(2/3)(0.5) = 0.167$. Without any manipulation threat $\mathcal{P}$ would be 5.075: the threat costs investors $0.5 \times 0.6 = 0.3$ but no NPV once the parachute is used.

Step 6 — The r-type does not mimic. Truthful, she gets $z^r r_H b$; claiming poor prospects yields at most $T^q = z^r(q_H - \tau)b < z^r r_H b$ when $q_H R < L$ ($0.6 < 2.85$ in the example), and $z^q r_H b \leq z^r r_H b$ when $q_H R > L$ ($z^q \leq 1 = z^r$). Her constraint is slack.

Answer . Optimal general contract: $R_b^r = B/\Delta p$, $T^r = 0$, $L_l^r = L$. If $q_H R > L$: no golden parachute, $T^{q*} = 0$ (continuation is the cheaper currency; $q_H R_b + dT/dz^q = q_H R - L > 0$); (NM) is met by raising z^q to θz^r, and under (7.10) funding fails whenever $(1, z^*)$ violates (NM). If $q_H R < L$: fire the low type, $z^{q*} = 0$, and pay $T^{q*} = z^r(q_H - \tau)B/\Delta p > 0$ (e.g. 0.6, raising NPV from 1.333 to 1.5). This confirms the heuristic of §7.2.1.2.

Intuition. Two currencies can buy honesty from a manager with bad news: keeping her job or paying her to leave. Keeping her is the cheaper currency when her continuation still creates value; when it destroys value, cash is cheaper, and the golden parachute is the price of truthful reporting.

Common mistakes.

- Calling golden parachutes pure waste: they buy incentive compatibility when continuation is inefficient.
- Treating the composition of the rent as important: investors care only about the expected rent Π; the parachute is needed only because continuation pay requires continuation.
- Forgetting that NPV is unaffected by transfers: only z^r, z^q change it.

Book answer. Matches the book (pp. 604–605): $R_b^r = B/\Delta p$, $T^r = 0$; when $q_H R > L$, $T^q = 0$; when $q_H R < L$, $z^{q*} = 0$ and $T^{q*} = z^r(q_H - \tau)B/\Delta p$; the r-type's (NM) does not bind. The book states (NM$'$) is binding; here it binds whenever the no-manipulation contract violates it (in case $q_H R > L$ with $\theta \leq z^*$ it is slack). The infeasibility result in Step 4 is an implication of (7.10) derived here, not stated in the book.

Ref: Tirole §7.2.1.1 "Managerial Myopia: The Incentive for Posturing" and §7.2.1.2 "Golden Parachutes", pp. 301–305; §3.2, pp. 115–117.